GENERAL SCIENCE GRADE 10: CHEMICAL BONDING

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.3: Chemical Bonding
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Bond and bond types (ionic, covalent, dative-covalent, hydrogen, metallic, Van der Waals) – Bond formation in common compounds and metals – Dot-and-cross diagrams for selected molecules and compounds (water, sodium chloride, hydrogen gas, oxygen gas, iodine) – Bond types and their structures (giant ionic, simple molecular, giant atomic, giant metallic) – Physical properties of elements and compounds as related to bond type (melting point, boiling point, solubility, electrical and thermal conductivity) – Structure and properties of Diamond – Structure and properties of Graphite – Structure and properties of Aluminium – Uses of Diamond, Graphite and Aluminium based on bond type and structure
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) illustrate chemical bonding in common compounds; b) identify bond types in common compounds; c) distinguish structures of selected elements and compounds; d) describe the relationship between the bond types and the physical properties of elements and compounds; e) select appropriate materials for use based on bond type and structure; f) reflect on the appropriateness of substances used in day-to-day life.
Core Competencies	– Communication and Collaboration: as the learner sensitises the community on use and care of common appliances such as aluminium utensils and other cookware, and discusses with peers the relationship between bond types and physical properties of different structures. – Digital Literacy: as the learner navigates through online sources to access virtual laboratories for simulations on formation of chemical bonds and researches bond types of common compounds. – Creativity and Imagination: as the learner undertakes a project of modelling ball-and-stick ionic and covalent structures of selected molecules using locally available resources such as maize cobs, clay and wire.
Core Values	– Respect: as the learner recognises and appreciates the input of every group member when discussing the relationship between bond types and physical properties of different structures.

	– Patriotism: as the learner sensitises the community on the appropriate use and care of common appliances such as aluminium sufurias and other cookware, contributing to responsible household management in Kenya. – Responsibility: as the learner observes safety precautions when carrying out activities to investigate properties of elements and compounds and handles laboratory apparatus with care.
Science & Engineering Practices	– Developing and Using Models: as the learner draws dot-and-cross diagrams and builds physical ball-and-stick models of ionic and covalent structures using locally available materials to represent bonding at the particle level. – Planning and Carrying Out Investigations: as the learner designs and conducts experiments to measure and compare physical properties (melting point, solubility, electrical conductivity) of substances with different bond types. – Analysing and Interpreting Data: as the learner collects experimental data on physical properties and constructs evidence-based explanations linking observed properties to underlying bond types and structures. – Constructing Explanations: as the learner explains why diamond is hard, graphite conducts electricity, and aluminium is malleable by reference to their bonding and structural type. – Obtaining, Evaluating and Communicating Information: as the learner navigates virtual laboratories and online resources to gather information on bond formation and communicates findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: as the learner observes safety precautions when carrying out activities to investigate properties of elements and compounds (e.g., handling hot apparatus when testing melting points, using electrical equipment safely). – Financial Literacy: as the learner makes informed decisions when selecting appropriate items, appliances and equipment made from specific elements and compounds — for example, choosing aluminium sufurias over less suitable alternatives based on thermal conductivity and cost.
Career Connections	– Materials Engineer / Metallurgist: understanding metallic bonding and the properties of aluminium and other metals underpins careers in Kenya's manufacturing and construction industries (e.g., Bamburi Cement, steel fabrication in Nairobi industrial area). – Chemical Technologist / Industrial Chemist: knowledge of ionic and covalent bonding is foundational for work in Kenya's chemical processing, fertiliser and pharmaceutical industries. – Jeweller / Gemologist: understanding giant atomic structures such as diamond and the properties that arise from covalent network bonding is directly relevant to gemstone trading and jewellery making. – Electrical Engineer / Electronics Technician: the relationship between bond type and electrical conductivity (metallic bonding in copper and aluminium wiring, graphite in electrodes) is central to electrical engineering work. – Environmental Health Officer: understanding how the bonding and solubility of compounds affects their behaviour in water and soil informs environmental monitoring and public health work around Lake Victoria and the Rift Valley.
Focus for Lessons	This unit investigates how atoms combine through different types of chemical bonds and how those bonds determine the observable physical properties of substances encountered in everyday Kenyan life — from the aluminium sufuria used to cook ugali, to the graphite pencil used in school, to the table salt (sodium chloride) sprinkled on nyama choma. Learners move from atomic-level models of bond formation (dot-and-cross diagrams and virtual simulations) to macroscopic investigations of melting point, solubility and conductivity, building a coherent, evidence-based explanation of structure–property relationships. The unit culminates in a community-facing project in which learners apply their understanding to evaluate and communicate the appropriate selection and care of household materials.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently — and what is happening between the atoms inside each one that controls every property we can measure?

	KICD KEY INQUIRY QUESTION: 1. How do bond types relate to physical properties of compounds?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Mysterious Behaviours of Three Kitchen Items: A teacher places three common items on the demonstration desk in front of the class: a lump of table salt (sodium chloride) harvested from the Kenyan coast, a short length of aluminium wire taken from a household sufuria handle, and a graphite rod removed from a used AA battery. The teacher then poses a series of puzzling observations for students to consider: (1) The table salt dissolves easily in a cup of water and, when the solution is connected to a simple circuit with a bulb, the bulb lights up — yet the dry salt crystals do not conduct electricity at all. The salt also has a surprisingly high melting point; it takes over 800 °C to melt it, far beyond what any home jiko could achieve. (2) The aluminium wire is shiny, bends without breaking, conducts electricity in the circuit immediately (even as a solid), and yet melts at a far lower temperature (660 °C) than the salt. (3) The graphite rod is dull grey, brittle (it snaps rather than bends), and yet it ALSO conducts electricity in the circuit — even though it is made entirely of carbon, a non-metal. A pencil leaves a mark because layers of graphite flake off easily onto paper. Students are asked: "All three of these substances are solids. Why do they behave so differently from each other when we test them for conductivity, melting point and mechanical strength? What is happening at the particle level inside each one that explains what we can see and measure?" This phenomenon is puzzling because learners' prior knowledge suggests that only metals conduct electricity, yet graphite — a non-metal — also conducts. It also puzzles learners that salt, which they see dissolve and conduct in solution, refuses to conduct as a solid. The phenomenon motivates the entire unit: the answer lies in the type of chemical bond holding the particles together and the resulting three-dimensional structure.
Storyline Thread	Lesson 1 — PREDICT: Anchoring Phenomenon & Prior Knowledge Elicitation Students observe the three demonstration items (salt crystal, aluminium wire, graphite rod) and record predictions about conductivity, melting point and hardness before any testing. Teacher facilitates a Predict phase: students write individual predictions, then share in pairs, then cold-call three groups to voice predictions and record them on the class Driving Question Board (DQB). Teacher introduces the driving question and activates prior knowledge of valence electrons and electron arrangement from Sub-Strand 2.1, asking: "What do you already know about how atoms share or transfer electrons?" Lesson 2 — OBSERVE & EXPLAIN: Valence Electrons and Bond Formation Students revisit dot-and-cross diagrams from Sub-Strand 2.1 and are guided to discuss the role of valence electrons in bond formation. Teacher demonstrates dot-and-cross diagrams for ionic bond formation (Na^+Cl^-) and covalent bond formation (H_2, H_2O, O_2) on the board. Learners practise drawing their own diagrams for assigned molecules in pairs. Introduction of the six bond types (ionic, covalent, dative-covalent, hydrogen, metallic, Van der Waals) with Kenyan-context examples for each. Students update the DQB with new questions arising. Lesson 3 — OBSERVE: Virtual Lab Simulation & Dot-Cross Practice

	Learners navigate online virtual laboratory simulations (or teacher-led projected simulation where devices are limited) to observe the formation of ionic and covalent bonds at the particle level. Students complete a structured worksheet drawing dot-and-cross diagrams for NaCl, MgO, HCl, NH_3 and CH_4. Pairs compare diagrams and identify patterns in electron transfer vs. sharing. Teacher cold-calls four pairs to present diagrams on the board and facilitates class correction. Students add observations to the DQB. Lesson 4 — OBSERVE: Structures of Elements and Compounds Teacher introduces the four structural categories: giant ionic, simple molecular, giant atomic and giant metallic. Learners use print/digital media to research the structures of NaCl (giant ionic), iodine (simple molecular), diamond and graphite (giant atomic/layered covalent) and aluminium (giant metallic). Students sketch structural diagrams in their notebooks and label the particles and forces. Small groups (4 learners) are assigned one structure each and prepare a 2-minute explanation to share with the class — jigsaw strategy. Lesson 5 — OBSERVE: Practical Investigation of Physical Properties Learners carry out a structured practical: testing solubility in water, electrical conductivity (simple circuit with bulb/LED), and observing relative hardness/brittleness for: NaCl (table salt), candle wax (simple molecular), graphite rod, and aluminium foil strip. Results are recorded in a class data table. Teacher facilitates whole-class data sharing and asks: "Which pattern in your data surprises you most?" — wait time 30 seconds, cold-call three students. Students connect experimental observations back to the anchoring phenomenon. Lesson 6 — EXPLAIN: Bond Types → Physical Properties (Sensemaking) Teacher leads structured sensemaking discussion: for each structural category, learners articulate why the bonding arrangement produces the observed physical properties (e.g., giant ionic — high melting point because many strong electrostatic attractions must be overcome; simple molecular — low melting point because only weak Van der Waals/hydrogen bonds between molecules; metallic — conducts because delocalised electrons move freely; giant atomic — very hard because every atom is covalently bonded in 3D network). Learners construct an evidence-based comparison table in their notebooks linking bond type → structure → property → everyday Kenyan example (sufuria, pencil, salt, candle wax). DQB is updated: "Which of our original questions can we now answer?" Lesson 7 — EXPLAIN & MODEL BUILDING: Diamond, Graphite and Aluminium Deep Dive + Community Project Learners explore the contrasting properties of diamond and graphite (both pure carbon) as a capstone explanation challenge. Teacher poses: "Same element, same type of bond — so why is one the hardest natural substance and the other slippery enough to use as a lubricant or electrode?" Groups discuss and present explanations referencing 3D tetrahedral vs. layered hexagonal structures and the presence of delocalised electrons in graphite. Learners then begin the community sensitisation project: drafting advice for a Kenyan household on the correct use and care of aluminium sufurias, graphite electrodes in torches, and NaCl in food and water treatment, grounded in bond-type reasoning. Lesson 8 — MODEL BUILDING & DQB REVIEW: Model Revision, Project Presentation & Unit Consolidation Learners build or refine physical ball-and-stick models of assigned molecules/structures using locally available materials (maize cobs as atoms, wire/string as bonds, clay balls colour-coded by element — echoing materials used by Kenyan primary-school teachers). Groups present their models and explain how the bonding determines at least two physical properties. Class revisits the original DQB: teacher facilitates a structured review — "Which questions did our evidence answer? What new questions remain?" Students write a final individual explanation connecting the anchoring phenomenon (salt, aluminium, graphite) back to bond types and structures. Teacher administers a short written exit task aligned to the KICD assessment rubric.

Supporting Phenomena	– Why does an ice cube (water, H_2O) melt in your hand at room temperature, while a piece of table salt (NaCl) from Mombasa would never melt on a hot day in Nairobi? — introduces the contrast between simple molecular and giant ionic structures. – Why can a Kenyan blacksmith in Kamuthe beat a piece of iron into a new shape without it shattering, yet a piece of dry ice (solid CO_2) crumbles and evaporates? — introduces metallic bonding and the concept of delocalised electrons allowing layers to slide versus the discrete molecules of a simple molecular solid. – Why does iodine (used in labs as an antiseptic indicator) sublime — turning from a shiny purple solid directly into a purple gas — when gently warmed, releasing a strong smell, while diamond (carbon) does not melt even in the hottest furnace? — contrasts Van der Waals forces in simple molecular solids with covalent network bonding in giant atomic structures. – Why do Kenyan electricians always choose copper or aluminium wire — never wood or plastic — to carry current in household wiring, and why can aluminium foil be pressed into any shape to wrap leftover ugali? — reinforces metallic bonding (delocalised electrons, mobile layers) as the structural explanation for conductivity and malleability.

LESSON 1 (40 minutes): The Three Kitchen Mysteries: Anchoring the Phenomenon & Eliciting Prior Knowledge	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit by anchoring learners in a puzzling, observable phenomenon drawn from everyday Kenyan kitchen items, and to surface and document prior knowledge and misconceptions that will drive inquiry across the entire sub-strand.
Knowledge	Learners will recall prior knowledge of particle structure of matter, basic electrical conductivity, and physical properties (melting point, hardness, brittleness) as a foundation for explaining why table salt, aluminium (sufuria), and graphite (pencil/battery rod) behave so differently as solids.
Skills	Learners will practise scientific observation, prediction-making with justification, questioning, and collaborative discussion as they engage with the anchoring phenomenon. They will begin constructing a Driving Question Board (DQB) to track the questions the phenomenon generates.
Attitudes	Learners will appreciate that everyday Kenyan kitchen objects are powerful scientific puzzles, and will develop curiosity and intellectual humility by recognising the limits of their current explanations.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	This is the opening lesson of the unit. Its sole purpose is to LAUNCH the anchoring phenomenon — the mysterious contrasting behaviours of table salt (from the Kenyan coast), an aluminium sufuria wire, and a graphite battery rod — so that every learner has a shared, concrete, puzzling experience to return to in every subsequent lesson. No explanations are given today. Instead, learners make predictions, surface misconceptions, generate questions, and open the Driving Question Board. The phenomenon creates the need-to-know that motivates Lessons 2–8.
Safety Notes	The simple conductivity circuit uses a 1.5 V AA battery or a 4.5 V flat battery — safe for classroom use. No mains electricity. Warn learners not to touch bare wire ends simultaneously. The graphite rod is from a used AA battery; handle with clean hands and wash after the lesson to avoid carbon dust contact with eyes. No heating or chemicals are used in this lesson.

B. LESSON OVERVIEW

In this 40-minute opening lesson, the teacher places three common Kenyan kitchen/household items — coastal table salt (NaCl), a length of aluminium wire from a sufuria handle, and a graphite rod from a used AA battery — on the demonstration desk and uses a simple 1.5 V battery-and-bulb conductivity circuit to reveal a series of deeply puzzling observations. Learners first predict in writing (individually), then share predictions in pairs, then witness the live demonstration. The class then co-constructs the opening Driving Question Board by contributing questions they cannot yet answer. No explanations are provided. The lesson closes with learners writing a personal 'best current explanation' that will be revisited and revised in every subsequent lesson. This lesson enacts the PREDICT phase of the ARES POE–DQB–Model cycle and establishes the intellectual need that drives the entire unit on Chemical Bonding.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner receives or copies	Teacher places the three items on	Individual Written Prediction	Teacher circulates and reads

📹 **VIDEO:** [Anatomy of a skeletal muscle cell](#) Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos](#) 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 [🔍 Search ARES for similar readings](#)	the 'Three Kitchen Items Prediction Sheet'. They see the three items displayed on the desk (salt in a small bowl, aluminium wire, graphite rod) but the conductivity circuit has NOT yet been switched on. Learners individually and silently write predictions for three test questions: (1) Which item(s) will light the bulb when placed in the circuit? (2) Which item do you think has the highest melting point — salt, aluminium, or graphite? (3) Which item do you think will snap rather than bend? Learners must write a reason (using particle-level language if they can) for each prediction. No discussion is allowed during this phase.	the desk and says: 'Before I switch anything on, I want to know what YOU think. On your paper, answer the three questions. You have 4 minutes — work silently and alone. There are NO wrong answers right now; I want your honest best thinking.' [START TIMER — 4 minutes silent writing.] While learners write, the teacher circulates and reads over shoulders, noting 3–4 interesting or divergent predictions without commenting. At the end of 4 minutes, the teacher says: 'Underline the one prediction you are LEAST confident about. That uncertainty is exactly where good science begins.'	(Predict-before-Observe). This strategy externalises prior knowledge and misconceptions before instruction, giving the teacher diagnostic data and giving learners a personal stake in the outcome of the demonstration. By committing predictions to paper, learners are cognitively primed to notice surprises during the observation phase.	prediction sheets. Observable indicators: (a) Does the learner predict that graphite will conduct? (Most will say NO — this is the target misconception to surface.) (b) Does the learner give a particle-level reason (e.g., 'metals have free electrons') or only a surface reason ('salt is white so it won't conduct')? (c) Does any learner correctly predict that dry salt will NOT conduct but dissolved salt will? Teacher mentally notes 2–3 learners with interesting predictions for cold-calling in Phase 2.
Observe Phase 📹 **VIDEO:** [Anatomy of a skeletal muscle cell](#) Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos](#) 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 [🔍 Search ARES for similar readings](#)	Learners turn to their neighbour and spend 3 minutes comparing predictions. Each pair must identify: (a) one prediction they AGREE on, and (b) one prediction where they DISAGREE. The teacher then calls on 5 pairs (cold-call) to share one agreement and one disagreement with the class. Predictions are recorded on a visible T-chart on the board labelled 'We PREDICT…' — no corrections are made, all ideas are honoured.	Teacher says: 'Turn to the person next to you. You have 3 minutes: find ONE thing you both agree on and ONE thing you disagree about.' [WAIT — 3 minutes pair talk.] Teacher then cold-calls exactly 5 pairs in sequence: 'Pair near the window — share your agreement.' [WAIT 10 seconds after naming pair before accepting answer.] After each pair responds, teacher writes their prediction verbatim on the T-chart saying: 'I am writing exactly what you said — no changes.' If a pair says 'We both think graphite won't conduct because it's not a metal,' teacher responds warmly: 'Interesting — you are in good company with many scientists who were also surprised. Let's keep that on the board and test it.' Teacher does NOT confirm or deny any prediction.	Think-Pair-Share with a Prediction Gallery. The public recording of predictions (including misconceptions) on the board creates a shared class record that will be visibly revisited and revised as evidence accumulates across lessons. This strategy builds psychological safety — learners see that all predictions are valued — and creates cognitive dissonance when the demonstration contradicts them.	Teacher listens for: (a) the prevalence of the misconception 'only metals conduct' (expected: ~80% of class will hold this view for graphite); (b) whether any learner distinguishes between dry salt and salt solution for conductivity (this is a high-value prior-knowledge indicator); (c) whether learners use the word 'electrons', 'ions', or 'particles' unprompted. Teacher makes a quick tally on a clipboard — this tally informs the depth of explanation needed in Lesson 3 (Ionic Bonding).
Explain Phase 📹 **VIDEO:** [Anatomy of a](#)	The teacher conducts the live demonstration in four steps, pausing after each for learners to	Teacher narrates each step in a calm, deliberate voice. Before STEP A: 'I am placing the	Discrepant Event Demonstration (Observe phase of POE). The deliberate sequencing — from	Teacher scans the room after STEP D and looks for: (a) facial expressions of genuine surprise at

skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	record observations on their sheet: STEP A — Aluminium wire in circuit: bulb lights immediately. STEP B — Dry salt crystals in circuit: bulb does NOT light. STEP C — Salt dissolved in a cup of water (using water from the teacher's flask): bulb lights. STEP D — Graphite rod in circuit: bulb lights. After all four steps, learners compare observations to their predictions and circle any prediction that was WRONG. The teacher then asks: 'Hold up your paper — how many of you had at least ONE prediction that was wrong?' (Show of hands, not shame — teacher raises their own hand too.)	aluminium sufuria wire in the circuit. Watch the bulb — say nothing yet, just observe.' [Connects wire. Bulb lights.] 'Write what you see.' [10-second WAIT.] Before STEP B: 'Now — dry table salt from the Kenyan coast, harvested near Malindi. Watch.' [Bulb does NOT light.] 'Write.' [10-second WAIT.] Before STEP C: 'Now I dissolve the same salt in water.' [Stirs. Connects electrodes to solution.] 'Same salt. Different result. Write what you see.' [10-second WAIT.] Before STEP D — PAUSE before connecting: 'This is a graphite rod from an old AA battery. Graphite is carbon. Carbon is a non-metal. Before I connect it — whisper to your neighbour: will it light the bulb?' [5-second whisper.] [Connects rod. Bulb lights.] 'Write.' [15-second WAIT — longer, to let surprise register.] Teacher then says: 'Show of hands — how many of you were wrong about at least one item?' [Raises own hand.] 'Good. Being wrong in science means we have something new to learn today.'	expected result (aluminium conducts) → surprising result (dry salt does NOT conduct) → more surprising result (dissolved salt DOES conduct) → most surprising result (graphite conducts) — escalates cognitive dissonance progressively. Each pause with written recording prevents the 'blink and miss' effect and forces deliberate noticing. The graphite result is the peak discrepant event because it most directly challenges the misconception 'only metals conduct electricity.'	graphite (a useful proxy for the misconception being activated); (b) learners who write 'I don't understand' or leave the observation blank (flag for support); (c) any learner who immediately reaches for a particle-level explanation ('maybe graphite has loose electrons?') — these are 'ready for deeper thinking' and can be used as peer explainers in Lesson 3. Teacher asks ONE cold-call question after all steps: 'Akinyi — what is the most surprising result to you, and why?' [WAIT 12 seconds before accepting answer.]
Driving Question Board (DQB) Creation 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding	Each learner receives one sticky note (or tears a small piece of paper). They write ONE question that the demonstration has raised for them — a question they cannot yet answer — beginning with 'Why…', 'How…', or 'What…'. Learners post their notes on the class DQB (a section of the board or a large sheet of manila paper headed: 'What are we trying to figure out?'). The teacher reads several aloud and groups them into emerging clusters (learners help by shouting 'same!' when they hear a question similar to theirs).	Teacher hands out sticky notes and says: 'You have seen something puzzling. Write ONE question — just one — that you cannot answer yet. Start with Why, How, or What. You have 2 minutes.' [WAIT — 2 minutes.] 'Now walk up and stick your question on the board.' [Allow 1 minute for posting.] Teacher then reads 6–8 notes aloud verbatim and says: 'Listen — if you hear a question similar to yours, say SAME.' Teacher uses this to cluster questions physically on the board. Teacher narrates as they	Driving Question Board (DQB) Co-Construction. The DQB is a living public record of the class's collective curiosity. By clustering learner-generated questions, the teacher makes the unit's learning trajectory transparent — learners can see what they will be figuring out and why. This strategy builds intrinsic motivation (these are OUR questions), provides the teacher with a formative snapshot of the class's collective understanding, and creates a reference point for the 'DQB Update' segment in every subsequent lesson.	Teacher reads all DQB notes before the next lesson to identify: (a) whether learners are asking particle-level questions ('What holds the salt together inside?') vs. only observable questions ('Why does salt dissolve?') — the ratio indicates readiness for sub-microscopic thinking; (b) whether any learner has already intuited the key idea ('Is it something to do with electrons being shared or given away?') — these learners can be stretched; (c) whether any cluster is absent (e.g., no questions about melting point) — if

[(Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Three anchor clusters are named by the class: (1) Questions about CONDUCTIVITY, (2) Questions about MELTING POINT, (3) Questions about STRUCTURE inside the solid.	cluster: 'It seems many of us are puzzled about WHY graphite conducts — that goes in the conductivity cluster. Several of you are asking what is INSIDE each solid — that goes in the structure cluster. These clusters are our Driving Question Board. Every lesson from now will answer at least one question on this board, and we will add new questions as we learn more.' Teacher takes a photo of the DQB (or assigns a learner to draw/copy it) for reference in future lessons.		so, the teacher asks a seeding question in the next lesson to open that gap.
Model Building Phase 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their seats and spend the final 5 minutes drawing or writing their 'Initial Model' — a best current explanation for WHY the three items behave differently. They may use words, diagrams, arrows, or particle drawings. The prompt on the board reads: 'Inside each solid, what do you think the particles look like and how are they held together? Draw or describe your idea for ONE of the three items.' Learners are told explicitly: 'This model does not need to be correct. It needs to show your honest best thinking TODAY. We will revise it every lesson until the end of the unit, and by the end you will be amazed how much it has changed.' Learners label their model 'Lesson 1 — Before Learning' and keep it in their science journal.	Teacher writes the model prompt on the board and says: 'In your science journal, draw or write your best current explanation for what is happening INSIDE one of these three solids. You have 5 minutes. Do not copy from a neighbour — I want YOUR thinking, even if it feels incomplete.' [WAIT — 5 minutes.] With 1 minute remaining: 'Add a label that says Lesson 1 — Before Learning and today's date. Keep this page — you will be amazed what you add to it by the end of this unit.' Teacher cold-calls 2 learners to briefly share (verbally) what they drew: 'Kamau — in one sentence, what does your model show?' [WAIT 10 seconds.] 'Wanjiku — same question.' Teacher responds to both with: 'Thank you — I can see your thinking. Hold that idea — we will test it very soon.' Teacher does NOT correct the models.	Initial Model Construction (Model-Based Reasoning). Asking learners to externalise their mental model before instruction creates a personal baseline that makes learning visible over time. The explicit instruction to keep and date the model — and the promise that it will change — builds a growth mindset around scientific modelling. It also gives the teacher a rich diagnostic of the class's sub-microscopic thinking (particle drawings, use of forces/bonds, structure representations) before any formal instruction begins.	Teacher collects or photographs 5–6 initial models (a sample across ability ranges) before the next lesson. Observable indicators assessed: (a) Does the learner draw particles at all, or only describe bulk properties? (b) Does any learner spontaneously draw a lattice or repeating structure for salt? (c) Does any learner use the word 'bond', 'electron', 'ion', or 'force' in their model? (d) Does the learner distinguish between the three items, or treat them as the same? These indicators set the entry point for Lesson 2 (Introduction to Bond Types).

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal: (1) Which item in the demonstration produced the greatest surprise — was it the graphite conducting, or the dry salt NOT conducting? What does this tell you about the dominant prior-knowledge framework in your class? (2) How many learners had at least one correct

prediction? Did any learner predict ALL four outcomes correctly — and if so, what prior knowledge are they drawing on? (3) Read through the DQB sticky notes: are your learners asking particle-level questions or only observable-level questions? If they are stuck at the observable level, plan a brief 'zoom in' prompt for the start of Lesson 2 ('Imagine you could shrink to the size of an atom and walk around inside the salt crystal — what would you see?'). (4) Were all learners able to draw SOMETHING for their initial model, or did some leave the page blank? Blank pages signal low confidence in particle thinking — these learners need explicit permission to use rough sketches and approximations. (5) Did the 40-minute period feel rushed? If so, Phase 2 (Pair Share) can be shortened to 2 minutes in future, or Phase 5 can be set as a brief homework task in the science journal.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed a live conductivity demonstration using a 1.5 V battery-and-bulb circuit with three Kenyan household items: table salt (from the coast), aluminium sufuria wire, and a graphite rod from a used AA battery. They saw that aluminium conducted immediately, dry salt did not conduct, dissolved salt did conduct, and — most surprisingly — graphite also conducted electricity even though it is a non-metal.
What did I learn?	Students surfaced and documented their prior knowledge and misconceptions about electrical conductivity, melting points, and mechanical properties of solids. They learned that their existing rule ('only metals conduct') is insufficient to explain all the evidence, and they identified specific gaps in their understanding that the unit will address.
How does this explain the phenomenon?	No explanations of bond types or particle structure were provided in this lesson. The lesson's purpose was to open the inquiry, not close it. The three items remain mysterious. Students have committed their initial predictions and initial models to their science journals — these will be revised and explained across Lessons 2–8 as each bond type (ionic, metallic, covalent/giant covalent) is introduced.

LESSON 2 (80 minutes (2 × 40-minute lessons)): Predicting the Invisible: What Are Our Three Kitchen Items Made Of?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To activate prior knowledge of valence electrons and electron arrangement, and to generate testable, evidence-based predictions about the conductivity, melting point and mechanical properties of table salt, aluminium wire and graphite — creating a productive knowledge gap that motivates the entire bonding inquiry.
Knowledge	Learners will recall that atoms have valence electrons (from Sub-Strand 2.1); that elements can lose, gain or share electrons to achieve stability; and that these electron interactions are the foundation of all chemical bonding. Learners will also know that solids, liquids and gases differ in particle arrangement.
Skills	Learners will practise: (i) writing structured individual predictions using an 'I think … because …' sentence frame; (ii) sharing and critiquing predictions in pairs using accountable talk; (iii) recording class predictions on a Driving Question Board (DQB); and (iv) connecting macroscopic observations (what we can see and measure) to particle-level thinking (what atoms might be doing).
Attitudes	Learners will appreciate that scientific thinking begins with honest uncertainty — that not knowing is a starting point, not a failure. They will value the contributions of every group member and listen respectfully to predictions that differ from their own.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	Lesson 1 introduced the anchoring phenomenon — the three kitchen items (salt, aluminium sufuria wire, graphite rod) and their puzzling behaviours. Lesson 2 is the formal PREDICT phase of the 5-phase learning sequence. Before any testing or instruction, learners must commit their prior-knowledge reasoning to the record. This creates a cognitive stake in the outcome: when evidence arrives in Lessons 3–6, learners will be able to compare what they predicted against what they observe, driving genuine sensemaking. Lesson 2 also surfaces misconceptions (e.g., 'only metals conduct electricity') that the teacher can explicitly address later.
Safety Notes	No chemicals or electrical equipment are used in this lesson. The aluminium wire, salt crystals and graphite rod are handled as display items only — learners should not break the graphite rod or taste the salt during this lesson. Remind learners that the salt on display may have handling contamination; it is not food-grade for tasting.

B. LESSON OVERVIEW

Learners enter to find the three anchoring-phenomenon items (table salt lump, aluminium wire from a sufuria handle, graphite rod from a used AA battery) displayed on the front desk — exactly as in Lesson 1, so the visual anchor is re-established immediately. The lesson runs entirely as a structured PREDICT phase. Learners first write individual 'I think … because …' predictions for three properties (electrical conductivity, melting point, mechanical strength/hardness) across all three items, drawing only on their Sub-Strand 2.1 knowledge of valence electrons and electron arrangement. Pairs then compare predictions and identify agreements and disagreements. The teacher cold-calls six pairs and records every prediction — agreements, disagreements and uncertainties — on the class Driving Question Board (DQB), which is displayed prominently and will remain on the classroom wall for the full eight-lesson sequence. The lesson closes by formalising the driving question and having learners articulate, in writing, the one thing they are most curious or confused about. This 'wonder statement' feeds directly into the DQB and primes Lesson 3's observation activities.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners enter and immediately see the three items on the front desk, labelled with small cards: 'Chumvi ya Pwani (NaCl) — harvested near Malindi', 'Wire from aluminium sufuria handle', 'Graphite rod from used AA battery — Nairobi'. Learners spend 2 minutes silently re-examining the items (no touching yet) and reading the question projected on the board: 'All three are solids. But are they the same KIND of solid? What clues do you have from what you already know about atoms and electrons?' Teacher does NOT explain or introduce new information. After 2 minutes, teacher asks learners to open their science journals to a fresh page titled: 'My Predictions — Lesson 2'.	As learners settle, teacher circulates silently, making eye contact and nodding — signalling that quiet observation time is intentional and valued. After 2 minutes, teacher says: 'Before we do a single test, before I tell you anything new, I want to know what YOU think. Scientists always predict before they observe — that is part of the science and engineering practice of asking questions and defining problems. So today, your job is to think hard and commit to a prediction.' Teacher then reads aloud the three properties learners will predict: '(1) Electrical conductivity — does electricity pass through it? (2) Melting point — how much heat does it take to melt it? High, medium or low compared to the others? (3) Mechanical strength — is it hard and rigid, or can it bend or snap?' Teacher writes these three properties on the board in a 3 × 3 grid (rows = properties, columns = three items). WAIT TIME: teacher pauses for 10 seconds after reading the grid to allow learners to process before giving the next instruction.	Re-anchoring to phenomenon (visual and conceptual). By placing the same three objects in view at the start of every lesson, the teacher ensures the phenomenon remains the cognitive reference point — every prediction learners write is about these specific, real, Kenyan objects, not abstract chemistry concepts. This prevents the lesson from feeling like disconnected theory.	Teacher observes whether learners look at the items thoughtfully or disengage during the silent observation period. Teacher notes any learner who immediately writes without looking — this may indicate rote recall rather than phenomenon-grounded thinking. Observable indicator: at least 80% of learners have their journals open and pencils poised within 3 minutes of entry.
Observe Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding	Learners individually complete a structured 'Prediction Table' in their science journals. The table has three rows (conductivity, melting point, mechanical strength) and three columns (NaCl salt, aluminium wire, graphite rod). For each of the nine cells, learners write: (a) their prediction (e.g., 'conducts / does not conduct'; 'high / medium / low'; 'hard and brittle / flexible / soft'), and (b) a one-sentence justification beginning	Teacher circulates continuously during the 12 minutes, reading over shoulders without commenting on correctness. Teacher uses two specific moves: (1) If a learner leaves a cell blank, teacher crouches, points to the cell and says quietly: 'What do you remember about how electrons behave in metals from Sub-Strand 2.1? Use that as your starting point.' WAIT TIME: 15 seconds of silence after prompting — do not	Individual prediction with sentence-frame scaffolding (NGSS Science and Engineering Practice 1: Asking Questions and Defining Problems). The 'I think … because …' frame forces learners to connect their macroscopic prediction (what will happen at the bench) to a particle-level reason (what atoms/electrons are doing). This is the first step in the explain-from-evidence cycle that runs through all 8 lessons. Scaffolding	Teacher collects a rapid sample: during circulation, teacher photographs (or mentally notes) the prediction tables of 4–5 learners to use as discussion anchors in Phase 3. Observable indicator: every learner has attempted all 9 cells with a justification before Phase 3 begins. Teacher notes the most common misconception for explicit addressing in Lesson 4 (Explain phase). Key misconception to

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	with 'because …' that references their knowledge of atoms or electrons. A sentence-frame scaffold is printed on a card on each desk: 'I predict that [item] will [property outcome] because I know that [particle/electron reasoning].' Learners work in silence for 12 minutes. They are explicitly told: 'There are no right or wrong answers yet — an honest prediction based on your current thinking is exactly what science needs.'	answer for the learner. (2) If a learner writes a justification that does not reference electrons or atoms (e.g., 'because it is shiny'), teacher says: 'Good observation — now go one level deeper. Why is it shiny? What are the electrons doing?' Teacher identifies 3–4 learners with particularly interesting — or particularly common misconception-revealing — predictions to cold-call in Phase 3. Teacher mentally flags: (a) any learner who predicts graphite will NOT conduct (very common misconception), (b) any learner who predicts salt WILL conduct as a solid (another common error), (c) any learner who gives a particle-level justification unprompted (strong prior knowledge to celebrate). At the 10-minute mark, teacher gives a 2-minute warning: 'Two more minutes — make sure every cell has a prediction AND a because-statement.'	with sentence frames is especially important in multilingual Kenyan classrooms where learners may have the concept but struggle to express it in English.	track: 'Graphite is a non-metal, so it cannot conduct electricity.'
Explain Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners turn to their desk partner and spend 8 minutes comparing prediction tables. They use a structured 'Agree / Disagree / Unsure' protocol: for each of the 9 cells, they mark whether they agree, disagree, or are both unsure. Where they disagree, they must each give their 'because' reasoning and then record BOTH predictions — they do not have to reach consensus. The pair then identifies their single biggest disagreement (the prediction they feel most uncertain or divided about) and writes it as a question: 'We are not sure whether _______ because _______.' This question will be contributed to the DQB in Phase 4. The final 2 minutes of pair work are used to rehearse:	Teacher circulates and listens in to pair conversations. Teacher uses targeted questioning when pairs reach quick false consensus: 'You both agree that graphite won't conduct — but graphite is used in some electrical contacts in real Kenyan factories. Does that change your thinking at all?' (Do NOT explain why — just introduce the cognitive conflict.) WAIT TIME: after this prompt, teacher steps back for 15 seconds and observes whether the pair revises their thinking or defends their original prediction. Both responses are scientifically valuable at this stage. Teacher selects 6 pairs to cold-call in Phase 4 based on: (1) two pairs with the most common prediction (to represent the class	Structured academic controversy / productive disagreement. In Kenya's CBE framework, communication and collaboration is a core competency. By requiring pairs to record BOTH predictions when they disagree — rather than forcing consensus — the teacher models that scientific disagreement is not a problem to be resolved by social pressure but by evidence. This sets the epistemic culture for the entire unit.	Teacher listens for the quality of justifications during pair talk. Observable indicator: learners reference valence electrons, electron arrangement or particle structure in at least half their 'because' statements. Teacher notes pairs who rely entirely on macroscopic properties (colour, shininess, texture) without any particle-level reasoning — these pairs will need additional scaffolding in the Observe phase (Lesson 3). Teacher also notes pairs who spontaneously use the words 'ion', 'bond' or 'electron' — evidence of strong prior knowledge to build on.

	each pair selects a spokesperson and prepares a 30-second share using the frame: 'We agreed that … We disagreed about … Our biggest question is …'	consensus), (2) two pairs with an outlier prediction (to create productive disagreement), (3) two pairs whose 'biggest question' is especially close to the driving question. Teacher writes these 6 pair-names on a sticky note to manage cold-calling efficiently.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the full DQB: a large sheet of chart paper (or a section of the classroom wall) divided into four sections labelled: 'What we PREDICT', 'What we OBSERVE', 'What we can EXPLAIN', and 'What we are still WONDERING'. In this lesson, only the first and last sections are populated. Each pair has been given two sticky notes (different colours): one for their 'biggest agreed prediction' and one for their 'biggest question'. Pairs write neatly and post their sticky notes as the teacher cold-calls them. The teacher cold-calls all 6 pre-selected pairs in sequence, asking each spokesperson to read their sticky notes aloud before posting. After all 6 pairs have shared, the teacher facilitates a 5-minute whole-class look at the DQB: 'Look at all these predictions and questions. Which ones appear more than once? Which ones contradict each other?' Learners identify clusters by a show of hands. Teacher circles clustered sticky notes and labels the clusters: e.g., 'Cluster A: Does graphite conduct?', 'Cluster B: Why does salt need so much heat to melt?', 'Cluster C: What makes aluminium bendy?'	Cold-call protocol: Teacher calls pairs by name (not by hand-raising) using the pre-selected list. For each pair: 'Pair [names], please read your prediction sticky note first — then your question sticky note.' After the pair reads, teacher asks ONE follow-up question to the class: 'Does anyone have a prediction that AGREES with this pair? Raise your hand.' Count hands aloud: 'Seven hands — so that is a popular prediction. Does anyone have a DIFFERENT prediction for the same property?' Count again. This creates a live frequency distribution of class predictions that makes the stakes of the upcoming evidence-gathering obvious. After all 6 pairs have shared, teacher reads the driving question aloud slowly: 'Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently — and what is happening between the atoms inside each one that controls every property we can measure?' Teacher then says: 'Every prediction you have posted today is our best current answer to this question. In the next six lessons, we will gather evidence to find out who is right — and more importantly, WHY.' WAIT TIME: after reading the driving question, teacher pauses for 15 seconds of	Driving Question Board as a living epistemic artefact (NGSS Science and Engineering Practice 8: Obtaining, Evaluating and Communicating Information). The DQB externalises the class's collective knowledge state — it makes thinking visible. By physically dividing the board into 'Predict / Observe / Explain / Wonder', the teacher communicates the structure of scientific inquiry as a process that moves from uncertainty toward explanation. Learners can see at a glance how much of the board is still empty — creating a genuine motivation to fill it through evidence.	Teacher reads every prediction sticky note as it is posted. Observable indicators of quality: (1) Predictions are specific and measurable (e.g., 'salt will NOT conduct as a solid') rather than vague ('salt is different'). (2) Question sticky notes are genuine knowledge-gap questions (e.g., 'Why does graphite conduct if it is not a metal?') rather than procedure questions ('How do we test conductivity?'). Teacher uses a tally to count: how many of the 6 shared predictions include a particle-level justification (target: at least 3 of 6). Any pair whose question sticky note is a procedure question rather than a conceptual question is gently redirected: 'That is a great HOW question — can you also write a WHY question? Why MIGHT graphite conduct, even though it is a non-metal?'

		silence before moving on.		
Model Building Phase 📹 **VIDEO:** [Precipitation reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Teacher leads a 10-minute structured recall of Sub-Strand 2.1 knowledge using a 'What do we already know?' spider diagram drawn live on the board. The central node is 'Valence Electrons'. Teacher asks four targeted questions and builds the diagram from learner responses: (Q1) 'How many valence electrons does sodium (Na) have?' (Q2) 'How many does chlorine (Cl) have?' (Q3) 'What happens when sodium and chlorine come near each other in terms of electrons?' (Q4) 'Carbon has 4 valence electrons — what might carbon do with those electrons when it meets another carbon atom?' After the spider diagram is complete, learners individually write a 'Wonder Statement' in their journals: one sentence beginning 'I wonder whether the way atoms share or transfer electrons explains why [choose one of the three items] behaves the way it does, because …'. These statements are NOT shared publicly yet — they are personal commitments that learners will revisit in Lesson 7 (the Explain phase). Lesson closes with the teacher previewing Lesson 3: 'Next lesson, we will actually TEST the three items. You will observe their conductivity, melting point and hardness with your own eyes. Then we will come back to every prediction on this board and mark what was right, what was wrong — and ask WHY.'	Spider diagram Q&A: teacher uses cold-call (not hand-raise) for all four questions. For Q1 and Q2, teacher cold-calls two learners who appeared confident during pair work — these are 'retrieval warm-up' questions where success is likely and builds momentum. For Q3, teacher cold-calls a learner who wrote a particle-level justification during Phase 2 (flagged during circulation) — this surfaces strong thinking publicly. For Q4, teacher cold-calls a learner who predicted graphite would NOT conduct — the goal is to create a gentle cognitive conflict: if carbon can share 4 electrons with other carbons, could it create a structure that allows electricity to flow? Teacher does NOT answer this question. Instead: 'Hold that thought — that is exactly what Lesson 3 is going to show us.' WAIT TIME: 15 seconds after each cold-call question before teacher provides any hint or rephrasing. If a learner cannot answer after 15 seconds, teacher uses a 'phone a friend' move: 'Can anyone add to what [name] is thinking?' — ensuring the original learner still gets to hear and respond to the peer answer. For the Wonder Statement, teacher circulates and reads 4–5 statements, selecting one to read anonymously to the class as a model of strong scientific curiosity: 'I am going to read one wonder statement without saying whose it is — listen for what makes it a great science question.' Teacher then explicitly connects the wonder statement to the driving question before	Prior knowledge activation via teacher-facilitated spider diagram (NGSS Science and Engineering Practice 6: Constructing Explanations). By building the Sub-Strand 2.1 spider diagram from learner responses — not by presenting it as a completed diagram — the teacher honours what learners already know and positions that knowledge as a genuine resource for the bonding inquiry. The 'Wonder Statement' strategy (adapted from the ARES Evidence-Gathering sequence) gives every learner a personal stake in the upcoming observations, because they have committed in writing to what they are curious about. This dramatically increases engagement in Lesson 3.	Teacher reads wonder statements during the final circulation. Observable quality indicators: (1) The wonder statement names a specific item (not 'one of them' vaguely). (2) The 'because …' clause references valence electrons, electron sharing/transfer, or particle arrangement. (3) The statement is genuinely uncertain — it does not confidently assert an answer. Teacher uses the wonder statements as an exit-ticket proxy: any learner whose wonder statement is too vague or missing a 'because' clause is given one verbal prompt before the bell: 'Can you add one word about electrons to your because statement?' This provides a quick formative snapshot of which learners are ready for Lesson 3's observation activities and which may need additional scaffolding.

		dismissing.		

D. TEACHER REFLECTION
After this lesson, reflect on the following questions in your professional journal: 1. PREDICTION QUALITY: When you circulated during individual prediction writing, what proportion of learners gave particle-level justifications (referencing electrons or valence electrons) versus purely macroscopic justifications (referencing colour, weight or appearance)? If fewer than half gave particle-level justifications, consider beginning Lesson 3 with a 5-minute retrieval quiz on electron arrangement before moving to the observation activities. 2. THE GRAPHITE MISCONCEPTION: How many learners predicted that graphite would NOT conduct electricity? This is the most important misconception to track across the sequence. Record the number on a sticky note and attach it to your lesson plan — you will need this baseline to measure conceptual change by Lesson 6. If more than two-thirds of the class holds this misconception, plan to give extra WAIT TIME in Lesson 3 after learners observe graphite conducting. 3. DQB QUALITY: Read every sticky note posted on the DQB tonight. Sort them into: (a) predictions that are testable and specific, (b) predictions that are vague, (c) questions that are conceptual and connected to bonding, (d) questions that are procedural. Plan to address category (b) and (d) at the start of Lesson 3 by asking the authors to refine their sticky notes after observing the evidence. 4. COLD-CALL EQUITY: Did you cold-call learners across gender and seating zones? In Kenyan classrooms, learners seated at the back or along the sides are often under-called. Review your 6 cold-called pairs — if they were all from the front-centre, adjust your pre-selection strategy for Lesson 3. 5. WONDER STATEMENTS: Collect 4–5 strong wonder statements (with learner permission) and type them up to display alongside the DQB before Lesson 3. This signals that individual curiosity is valued and that the class's questions are driving the inquiry, not just the teacher's agenda.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Three kitchen items — a lump of Kenyan coastal salt (NaCl), a length of aluminium sufuria wire, and a graphite rod from a used AA battery — displayed on the demonstration desk. No testing was done in this lesson; learners observed the items visually and by recall only.
What did I learn?	Learners activated prior knowledge of valence electrons and electron arrangement from Sub-Strand 2.1. They practised writing structured predictions using an 'I think … because …' frame, connecting macroscopic properties (conductivity, melting point, mechanical strength) to particle-level reasoning about how atoms might share or transfer electrons.
How does this explain the phenomenon?	Nothing has been formally explained yet — this lesson is intentionally the PREDICT phase. All predictions remain on the Driving Question Board as hypotheses to be tested. The class driving question was formally introduced: 'Why do these three solids behave so differently, and what is happening between the atoms inside each one?'

LESSON 3 (80 minutes (2 × 40-minute lessons)): Ionic Bonding: Why Does Table Salt Conduct Only When Dissolved?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to explain how ionic bonds form through electron transfer, describe the giant ionic lattice structure of sodium chloride, and relate that structure to the properties observed in the anchoring phenomenon — high melting point, brittleness, and conductivity only in solution or melt.
Knowledge	Learners will be able to: (a) illustrate ionic bond formation using dot-and-cross diagrams for NaCl and MgO; (b) describe the giant ionic lattice as a three-dimensional array of alternating cations and anions; (c) explain why ionic solids do not conduct electricity in the solid state but do conduct when dissolved in water or melted; (d) explain why ionic compounds have high melting points and are brittle.
Skills	Learners will be able to: (i) draw dot-and-cross diagrams accurately for ionic compounds; (ii) construct a 2-D lattice model using locally available materials (bottle caps, clay balls, maize grains of two colours); (iii) analyse data from a conductivity test to draw evidence-based conclusions; (iv) communicate particle-level explanations using the claim–evidence–reasoning (CER) framework.
Attitudes	Learners will: appreciate that observable macroscopic properties of everyday substances such as coastal table salt are direct consequences of invisible atomic-level structures and interactions; show respect for peers' contributions during group model-building; demonstrate responsibility when handling electrical circuit apparatus.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	In Lesson 2, learners mapped valence electrons and identified that atoms seek a stable octet. This lesson completes the first branch of the storyline by explaining the ionic bond in NaCl — the salt from the demonstration desk. Learners will now be able to answer WHY the salt conducts only in solution (ions are locked in place as a solid; free to move when dissolved) and WHY it has such a high melting point (strong electrostatic forces throughout the entire lattice must all be broken simultaneously). This evidence directly advances the Driving Question Board entry: 'What holds the particles in salt together?' and sets up Lesson 4 (covalent bonding in graphite and simple molecules).
Safety Notes	LOW RISK. When using the conductivity circuit (9-V battery, two graphite electrodes, LED bulb, connecting wires, saline solution in a plastic cup): ensure no bare wire contacts learner hands; dry hands before touching connectors; do not allow the electrodes to touch each other directly (short circuit). Dispose of saline solution down the sink. If using clay or food dye for models, wash hands after the activity. No Bunsen burners or corrosive chemicals are used in this lesson.

B. LESSON OVERVIEW

Learners begin by predicting whether solid salt and dissolved salt will both light a bulb, recording atomic-level justifications using their Lesson 2 knowledge of valence electrons. They then gather evidence through a teacher-led conductivity demonstration (solid NaCl vs. NaCl solution vs. distilled water) and a dot-and-cross diagram activity. In the Explain phase, learners use the Claim–Evidence–Reasoning (CER) framework to construct a particle-level explanation for each observation, guided by a 2-D lattice diagram. Groups build physical lattice models from two-coloured maize grains or clay balls to represent Na^+ and Cl^- ions, then use the model to explain brittleness (layer shift brings like charges together → repulsion → cleavage). The lesson closes with DQB updates and a 3-minute exit ticket. Throughout, all observations are anchored to the three kitchen items on the demonstration desk.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Molecule Polarity Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work individually for 4 minutes, then share with a shoulder partner for 2 minutes. Each learner answers THREE written predict prompts in their science notebook: (1) 'Will dry salt crystals light the bulb? Draw the particles inside the solid crystal and use your drawing to explain your answer.' (2) 'Will salt dissolved in water from Lake Victoria light the bulb? Draw what you think happens to the Na and Cl particles when salt dissolves.' (3) 'From Lesson 2, you know Na has 1 valence electron and Cl has 7. Predict what happens to those electrons when Na and Cl come together — does one atom take the electron from the other, or do they share it?' Learners annotate their Lesson 2 dot diagrams (already in notebooks) to support their predictions. Pairs share one key idea with the class.	SETUP (before class): Place the three kitchen-item specimens back on the demonstration desk — coastal salt lump, aluminium wire, graphite rod — as a visual anchor. Write the three predict prompts on the board or display them on the ARES screen. SAY EXACTLY: 'Before we test anything today, I want your best scientific guess written down. A wrong prediction is not a failure — it is evidence your brain is working. You have 4 minutes of silent thinking time. Go.' [START TIMER — 4 minutes silent individual work.] After 4 minutes: 'Turn to your shoulder partner. Compare your particle drawings. Do your drawings agree on whether the Na and Cl particles are free to move in the solid? You have 2 minutes.' [CIRCULATE — scan notebooks for the key misconception: learners who draw NaCl as covalent (shared electrons) rather than ionic (transferred electron). Note 2–3 such notebooks to address during Observe.] After pairs share: cold-call 3 learners — select one who predicted 'solid will NOT conduct', one who predicted 'both will conduct', and one who is uncertain. SAY: 'We have three different predictions. Let us run the test and find out who the evidence agrees with.' [WAIT TIME: after each cold-call, pause 10 seconds before acknowledging, to allow other learners to evaluate the reasoning aloud.]	PREDICT-OBSERVE-EXPLAIN (POE) launch. By committing a particle-level prediction to paper BEFORE seeing evidence, learners activate prior knowledge (valence electrons from Lesson 2) and generate a cognitive conflict that motivates the Observe phase. The written prediction also serves as a personal baseline the learner can compare against during the Explain phase, making conceptual change visible and owned by the learner.	OBSERVABLE INDICATOR: Learner notebooks show a particle drawing for both solid and dissolved states (even if incorrect). Teacher scans for two key ideas — (a) whether learners represent NaCl particles as charged ions or as neutral atoms/molecules; (b) whether learners mention electron transfer from Na to Cl. Learners who draw neutral NaCl molecules (common misconception from covalent-first exposure) are flagged for targeted questioning during Phase 2. Do NOT correct at this stage — preserve the cognitive conflict.
Observe Phase	PART A — CONDUCTIVITY DEMONSTRATION (10 min): The	PART A — DEMONSTRATION SEQUENCE: Hold up the salt	EVIDENCE TABLE + WORKED EXAMPLE SCAFFOLDING. The	OBSERVABLE INDICATORS: (1) Evidence table has three rows

📹 **VIDEO:** [Video: Molecule Polarity](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	teacher runs three conductivity tests at the front bench while learners complete an evidence table in their notebooks with columns: 'Substance tested | Bulb lights? (Y/N) | What I think is happening to the particles'. Tests: (i) dry NaCl crystals between two graphite electrodes; (ii) distilled water; (iii) NaCl dissolved in tap water (approx. 10 g per 200 mL, prepared before class). Learners record each result immediately. PART B — DOT-AND-CROSS DIAGRAM ACTIVITY (10 min): Learners are given a printed or ARES-screen-displayed worked example of the dot-and-cross diagram for the formation of NaCl (Na donates 1 electron → Na^+; Cl accepts 1 electron → Cl^-). They then independently attempt the dot-and-cross diagram for MgO (Mg has 2 valence electrons; O has 6). They label: the electron being transferred, the resulting charges, and the electron configuration of each ion. Learners who finish early draw the lattice arrangement: alternating + and − charges in a 3×3 grid.	lump. SAY: 'This salt was harvested from the Kenyan coast near Malindi. Watch carefully — I am not explaining yet, I am just showing you evidence.' Test (i): Insert dry NaCl between electrodes. Pause — no glow. SAY EXACTLY: 'Record what you see. Do not explain yet.' [WAIT 10 seconds for recording.] Test (ii): Distilled water. Pause — no glow or very faint glow. SAY: 'Record.' Test (iii): NaCl solution. Bulb lights. SAY: 'Record. Now look back at your prediction. Put a tick or cross next to your prediction — were you right?' [WAIT 15 seconds.] Cold-call 2 learners: 'Amina, what did you predict and what happened?' then 'Kipchoge, your drawing showed the particles — do you think the particles in the solution are the same as in the solid crystal? Explain your thinking.' [WAIT 10 seconds after each.] PART B — DOT-AND-CROSS SCAFFOLDING: Display the NaCl worked example. Walk to the three learners flagged during Predict who drew covalent sharing. Crouch to eye level. SAY QUIETLY: 'Look at your Lesson 2 notebook — Na has how many valence electrons? Just one. Can it easily share that with Cl, or is it easier to give it away completely? Think about what a stable octet requires.' Do NOT give the answer — prompt only. After 6 minutes of independent work, call one volunteer to the board to draw the MgO dot-and-cross diagram. Whole-class check: 'Do you agree? Hold up thumbs-up or thumbs-down.' Address any errors immediately using the Socratic question: 'If Mg gives away 2 electrons, what charge does it now	evidence table structure (Substance | Observation | Particle interpretation) trains learners to separate empirical observation from theoretical inference — a core NGSS Science and Engineering Practice (SEP 4: Analysing and Interpreting Data). The worked NaCl dot-and-cross example provides a cognitive anchor for the novel MgO task; the peer board-check immediately surfaces and corrects errors before they are consolidated.	completed with correct Y/N for bulb — if distilled water row says 'Y', teacher re-demonstrates and clarifies. (2) MgO dot-and-cross diagram shows Mg^{2+} (2,8 configuration) and O^{2-} (2,8 configuration) with 2 electrons shown transferring. Common error to watch: learners draw only ONE electron transferring from Mg (giving Mg^+ and O^-) — address during whole-class board-check. (3) Thumbs-up/down check gives teacher real-time data on class understanding before moving to Explain.

		carry?'		
Explain Phase 📹 **VIDEO:** Video: Molecule Polarity Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — CER WRITING (8 min): Using their evidence table from Phase 2, learners individually write a Claim–Evidence–Reasoning (CER) statement answering: 'Why does solid NaCl NOT conduct electricity, but NaCl solution DOES?' Sentence starters are displayed: 'CLAIM: Solid NaCl does not conduct electricity because… / EVIDENCE: In our test, we observed that… / REASONING: This is because in the solid state, the Na^+ and Cl^- ions are… whereas in solution, the ions…' PART B — LATTICE MODEL BUILDING (12 min): Groups of 4 build a 2-D ionic lattice using two-coloured materials (e.g., yellow maize grains = Na^+; white maize grains = Cl^-, OR two colours of clay balls, OR two-colour bottle caps with + and − written on them). Each group arranges a 4×4 alternating grid on their desk. Teacher then poses two questions to the model: (i) 'Can any of your Na^+ or Cl^- particles move to carry charge? What would they have to overcome?' → groups physically try to slide one grain and observe that the whole structure resists. (ii) 'What happens if I push an entire row one position to the right?' → groups shift one row and observe that two yellow grains now sit next to each other (Na^+ next to Na^+) → repulsion → the model snaps/separates, explaining brittleness. Groups annotate their model with a sticky note: 'This shows why salt shatters instead of bending.'	PART A — CER LAUNCH: Display the CER sentence starters. SAY EXACTLY: 'A claim without evidence is just an opinion. A scientist always links what they say to what they saw. You have 8 minutes to write your CER. Use your evidence table from Phase 2 and your dot-and-cross diagram. Go.' [CIRCULATE — read over shoulders. Target learners who write 'because the salt dissolves' as reasoning — prompt: 'But WHY does dissolving allow conduction? What do the ions do that they could not do in the solid?'] After 8 minutes: cold-call 2 learners to read their CER aloud. Class evaluates: 'Does that CER include particle-level language — ions, lattice, movement? Thumbs up if yes.' Refine one CER together on the board as a class model. PART B — MODEL BUILDING: Distribute materials (pre-bagged: 8 yellow grains + 8 white grains per group). SAY: 'You are building the inside of this salt lump' — hold up the coastal salt specimen — 'at the particle level. Arrange your grains in an alternating pattern — no two of the same colour can touch.' [CIRCULATE — ensure alternating pattern. After 4 minutes:] SAY: 'Now I want you to try to slide one single grain out of the lattice. What happens to the forces?' [WAIT 15 seconds for groups to try.] SAY: 'Now shift the ENTIRE second row one position to the right. What do you notice about the charges that end up next to each other?' [WAIT 15 seconds.] Cold-call one group: 'Wanjiku, what happened when you shifted the row — and what does that mean for the real salt	CLAIM–EVIDENCE–REASONING (CER) FRAMEWORK + PHYSICAL ANALOGICAL MODEL. CER is an NGSS SEP 6 (Constructing Explanations) strategy that makes the logical structure of scientific argument explicit and visible to learners. The physical lattice model is an analogical modelling strategy (NGSS SEP 2): by manipulating a concrete representation, learners develop an embodied understanding of why the lattice is both strong (high melting point) and brittle (layer-shift repulsion) — properties that are otherwise abstract. Crucially, the model is connected back to the phenomenon (the salt lump on the desk) at every step.	OBSERVABLE INDICATORS: (1) CER statements contain particle-level language: 'ions', 'fixed positions', 'free to move'. Teacher collects 4 random CER notebooks at the end of Phase 3 to review during break. (2) Model shows correct alternating pattern — teacher photographs each group's model with a tablet/phone for the ARES platform upload. (3) Groups can verbally explain the row-shift brittleness mechanism when cold-called. RED FLAG: any group that cannot explain why like-charge neighbours cause repulsion needs a brief teacher revisit using the analogy: 'Two Maasai warriors facing each other — same side, they push each other away.'

		crystal when we hit it with a hammer?' [WAIT 10 seconds.] Connect explicitly to phenomenon: SAY: 'This is why the salt in our demonstration does not bend like the aluminium wire — it shatters. Your model just explained a property we can see with our own eyes.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Molecule Polarity] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner writes ONE new sticky note (or ARES digital card) to add to the class DQB under the column 'What we NOW understand'. The sticky note must answer one of the three sub-questions the class placed on the DQB in Lesson 1: (a) 'Why does salt only conduct when dissolved?'; (b) 'Why is the melting point of salt so high?'; (c) 'Why does salt shatter?' Learners also review the existing DQB cards from Lessons 1 and 2 and draw a line (using string or a drawn arrow on the board) connecting their new card to any earlier evidence or question it answers or relates to. The class identifies ONE remaining mystery: 'If ions explain salt, what explains the graphite rod — it has no ions at all, yet it ALSO conducts. What is different inside graphite?' This question is written in RED on the DQB as the bridge to Lesson 4.	Distribute sticky notes (or open ARES DQB tool). SAY EXACTLY: 'Look at the DQB. In Lesson 1, we asked three questions about salt. In Lesson 2, we learned about valence electrons and octets. Today, you have evidence and a model. Write ONE sticky note that answers one of those Lesson 1 salt questions — use particle language.' [GIVE 3 minutes for writing.] As learners post their notes, facilitate connections: 'Ochieng just posted a note about ions being locked in the lattice — does that connect to Fatuma's Lesson 2 note about electrons being transferred? Fatuma, come and draw a line between them.' After all notes are posted, SAY: 'We have explained the salt. Look at the graphite rod on the desk. Graphite has no Na, no Cl, no ions. It is pure carbon — all atoms the same. Yet it conducts. Our current model of ionic bonding CANNOT explain that.' Pause. 'What NEW question should we put on the DQB in red — the question we need to answer next?' Cold-call 2 learners. Write the agreed question in red marker: 'If there are no ions in graphite, how does it conduct electricity?' SAY: 'That is what Lesson 4 is for.'	DRIVING QUESTION BOARD (DQB) PROGRESSIVE REVEAL. The DQB is a collaborative sense-making tool aligned with NGSS SEP 1 (Asking Questions). By physically connecting new cards to previous evidence, learners make the epistemic structure of the unit visible — they can SEE how their understanding is growing across lessons. The deliberate placement of an unanswered RED question maintains productive intellectual tension and motivates the next lesson, rather than presenting science as a closed body of facts.	OBSERVABLE INDICATORS: Sticky notes contain at least one of the following particle-level terms: 'ions', 'lattice', 'electrostatic forces', 'fixed', 'free to move'. Teacher counts the proportion of cards that use particle language vs. macroscopic language only (e.g., 'because it dissolves'). Target: ≥75% of cards use particle language by end of Lesson 3. Cards using only macroscopic language are flagged for follow-up in Lesson 4 opener.
Model Building	Each learner individually updates the CUMULATIVE UNIT MODEL	SAY EXACTLY: 'Open your unit model — the split diagram you	CUMULATIVE MODEL REVISION (NGSS SEP 2: Developing and	EXIT TICKET ASSESSMENT: Teacher reviews exit tickets before

Phase				
📹 **VIDEO:** [Video: Molecule Polarity] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	in their science notebook. The unit model is a split diagram begun in Lesson 1 showing the three kitchen items. For the SALT section, learners now add: (i) a dot-and-cross diagram showing Na donating an electron to Cl; (ii) a small 3×3 lattice grid labelled 'Giant Ionic Lattice' with alternating Na^+ and Cl^-; (iii) three annotation arrows explaining: 'Strong electrostatic forces in all directions → HIGH melting point (801 °C)', 'Ions locked in fixed positions in solid → NO conductivity as solid', 'Ions free to move in solution → CONDUCTS electricity in solution'. Learners also complete a three-row summary comparison table begun in Lesson 2 — today they fill in the SALT row: Bond type | Particles present | Can conduct solid? | Can conduct in solution? | Melting point (high/low) | Malleable/brittle. The aluminium and graphite rows remain blank — to be completed in future lessons. This incomplete table is a deliberate scaffold to maintain the storyline.	started in Lesson 1. You are going to update ONLY the salt section today. The aluminium and graphite sections stay blank for now — we have not fully explained those yet.' [Display the model structure on the ARES screen as a reference.] [GIVE 10 minutes for individual model updating. CIRCULATE — check for: correct lattice alternation pattern; correct labels Na^+ and Cl^- (not Na and Cl); three annotation arrows present.] After 10 minutes: SAY: 'Hold up your model. Look at your neighbour's. Are the ions labelled with charges? If not, add the + and − now.' [GIVE 2 minutes for peer check.] Final 3 minutes — EXIT TICKET (written on a scrap of paper or the ARES platform): 'Mama Wanjiku in Nakuru dissolves salt in water to make githeri. She tests the solution with a battery and a bulb — it lights up. Using particle language, explain in 2 sentences why the solution conducts but the dry salt crystals in the packet do not.' Learners hand in the exit ticket as they leave. SAY: 'Your exit ticket is your ticket out of the door today.'	Using Models). The unit model is intentionally incomplete at every lesson — only the section relevant to the current lesson's evidence is filled in, with blank spaces acting as visual reminders of unresolved questions. This structure mirrors how scientific models are built iteratively from evidence, rather than presented as complete frameworks to be memorised. The comparison table adds a cross-cutting concepts lens (Structure and Function; Patterns) by asking learners to notice regularities across bond types as more rows are completed in later lessons.	Lesson 4. Success criterion: response mentions (a) ions locked/fixed in solid state and (b) ions free to move in solution/water. SCORING GUIDE — Strong (both a and b with particle language) / Developing (one of a or b, or both in macroscopic language) / Needs support (no particle language, or incorrect claim such as 'electrons move in solution'). Learners scoring 'Needs support' are paired with a 'Strong' peer at the start of Lesson 4 for a 3-minute partner re-teach before the new content begins.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) MISCONCEPTION TRACKING — How many exit tickets showed the misconception that 'electrons' (rather than ions) carry charge in the NaCl solution? If more than 30% of learners held this view, begin Lesson 4 with a 5-minute whole-class re-teach using the analogy of ugali porridge: 'When you stir salt into uji, the Na^+ and Cl^- particles separate and swim freely — it is the ions that move to the electrodes, not electrons jumping through the water.' (2) MODEL QUALITY — Did learners' cumulative unit models correctly label Na^+ and Cl^- (with charges) rather than Na and Cl? Unlabelled charges are a persistent gap — consider a quick 2-minute 'charge audit' at the start of Lesson 4 where learners check their own model against a projected exemplar. (3) DQB ENGAGEMENT — Did at least 75% of sticky notes use particle-level language? If not, the Explain phase CER activity may need more structured scaffolding in future lessons — consider pre-printing the CER sentence starters as a strip that learners paste into their notebooks rather than copying from the board. (4) KENYA CONTEXT — Did the reference to coastal salt (Malindi/Mombasa) and the githeri exit ticket make the chemistry feel relevant? Note any learner comments that connected the lesson to their home experience and use these as opening hooks for Lesson 4. (5) TIMING — The lattice model-building activity is the most time-intensive element. If the lesson ran over time, the DQB update can be shortened to 5 minutes by having learners write their sticky note AS their exit ticket and posting it digitally on ARES rather than physically on the board.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Dry NaCl crystals placed in a conductivity circuit did NOT light the bulb. Distilled water did NOT light the bulb. NaCl dissolved in water DID light the bulb. A dot-and-cross diagram showed the Na atom donating its single valence electron to the Cl atom, forming Na^+ and Cl^- ions with complete octets. A 2-D lattice model built from two-coloured maize grains showed alternating positive and negative ions in a regular pattern; shifting one row caused same-charge ions to become neighbours.
What did I learn?	Ionic bonds form by the complete transfer of one or more electrons from a metal atom to a non-metal atom, creating oppositely charged ions (cations and anions). In solid NaCl, Na^+ and Cl^- ions are held in a giant ionic lattice by strong electrostatic forces acting in all directions. These forces require very high temperatures to overcome, giving NaCl its high melting point (801 °C). In the solid state, ions are locked in fixed positions and cannot carry charge, so solid NaCl does not conduct electricity. When NaCl dissolves in water or is melted, the ions become free to move, allowing electrical conduction. The lattice structure also explains brittleness: a mechanical blow shifts an ion layer so that like-charged ions align, creating repulsive forces that split the crystal.
How does this explain the phenomenon?	The coastal table salt on our demonstration desk does not conduct electricity as a solid because its Na^+ and Cl^- ions are locked in fixed positions in a giant ionic lattice — they cannot move to carry charge. When the salt dissolves in water (as in a cup of githeri broth), the lattice breaks apart and the ions become free to migrate to the electrodes, completing the circuit and lighting the bulb. The same lattice — with its strong electrostatic forces acting throughout the entire three-dimensional structure — also explains why salt requires over 800 °C to melt, far beyond what a home jiko can achieve. The salt's brittleness (it shatters rather than bends) is explained by the layer-shift repulsion in the lattice model. This fully accounts for the salt observations in our anchoring phenomenon. The aluminium and graphite items on the desk still need a different type of bonding to explain their behaviour — that is the question now written in red on our Driving Question Board.

LESSON 4 (80 minutes): Valence Electrons, Dot-and-Cross Diagrams, and the Six Bond Types	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to use dot-and-cross diagrams to illustrate ionic and covalent bond formation, and to identify all six bond types with Kenyan-context examples — building the conceptual vocabulary needed to explain why the salt, aluminium sufuria, and graphite pencil from our kitchen table behave so differently.
Knowledge	Learners will know: (a) the role of valence electrons in chemical bond formation; (b) how to construct dot-and-cross diagrams for ionic bond formation (Na^+Cl^-) and covalent bond formation (H_2, H_2O, O_2); (c) the definitions and distinguishing features of the six bond types: ionic, covalent, dative-covalent, hydrogen, metallic, and Van der Waals forces.
Skills	Learners will be able to: (i) draw accurate dot-and-cross diagrams for assigned ionic and covalent molecules; (ii) classify a given substance into its correct bond-type category using evidence from its formula and properties; (iii) construct and update a class Driving Question Board (DQB) entry linking new bond-type knowledge to the anchoring phenomenon.
Attitudes	Learners will appreciate that the invisible arrangement of electrons between atoms determines every measurable property of everyday materials — from the coastal salt dissolving in water to the aluminium sufuria conducting electricity — and will value careful, evidence-based reasoning over surface-level observation.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	Lessons 1–3 established that the three kitchen items differ in conductivity, melting point, and mechanical behaviour. Lesson 4 now gives learners the shared language and diagrammatic tools (dot-and-cross diagrams and the six bond-type framework) that all subsequent lessons will use to explain those differences structurally. Without this lesson, learners cannot distinguish why Na^+Cl^- (ionic) behaves differently from Al (metallic) or graphite (covalent/Van der Waals), and the driving question cannot be resolved.
Safety Notes	No hazardous chemicals are used in this lesson. If a battery is opened to display the graphite rod, the teacher should wear latex gloves and ensure the battery is fully discharged. Broken graphite rods have sharp edges — handle with care and dispose of fragments in the bin. Learners must wash hands after handling any battery components. Digital device use must follow the school's cyber-safety guidelines.

B. LESSON OVERVIEW

In this 80-minute lesson, learners revisit dot-and-cross diagrams introduced in Sub-Strand 2.1 and deepen their understanding by applying them to bond formation. The teacher first elicits prior knowledge through a structured prediction task linked to the anchoring phenomenon — asking learners to sketch (from memory) what is happening between particles in each of the three kitchen items. A teacher-led demonstration then models dot-and-cross diagrams for Na^+Cl^- (ionic) and H_2, H_2O, O_2 (covalent) on the board, with explicit think-alouds connecting electron transfer/sharing to the behaviour of coastal salt and water. Learners then practise in pairs, drawing diagrams for assigned molecules. In the Explain phase, the teacher introduces all six bond types (ionic, covalent, dative-covalent, hydrogen, metallic, Van der Waals) using Kenyan-context anchors: coastal NaCl for ionic; water in a Nairobi household tap for hydrogen bonds; the aluminium sufuria for metallic bonds; iodine crystals used in lab tests for Van der Waals; the graphite rod for covalent/Van der Waals layering; and the dative bond in the ammonium ion formed when ammonia fertiliser (DAP used by Rift Valley farmers) reacts with an acid. The lesson closes with learners updating the class DQB and revising their particle-level models of the three kitchen items.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their notebook to a fresh page and draws a 3-column table headed 'Table Salt (NaCl)', 'Aluminium Sufuria Wire', and 'Graphite Rod'. Without discussion, they sketch their best guess of the particle-level picture inside each solid — showing atoms, any charges, and how they are held together. They write one sentence per column predicting: 'I think the particles are held together by…'. After 5 minutes of silent individual work, learners share predictions with a shoulder partner (2 minutes). Two or three pairs are cold-called to read their predictions aloud to the class. Teacher records key prediction words on the board (e.g., 'sharing', 'transfer', 'attraction', 'electrons moving') without correcting them yet.	1. Display the three physical items (or photographs) from the anchoring phenomenon on the demonstration desk. Say: 'Before we learn anything new today, I want to know what YOU already think is happening inside each of these solids at the particle level. You have studied electron arrangement in Sub-Strand 2.1 — use that knowledge now. You have 5 minutes. Work in silence.' START TIMER. 2. After 5 minutes, say: 'Turn to your shoulder partner. Compare your sketches. Do you agree or disagree? You have 2 minutes.' 3. After pair-share, say: 'I am going to cold-call three pairs. Tell me ONE word that describes how you think particles are held in the salt.' WAIT 10 SECONDS after asking. Cold-call Pair 1, then Pair 2, then Pair 3. Write their key words on the board under each column. 4. Do NOT confirm or deny any prediction. Say: 'Hold onto these ideas. We will revisit them at the end of today's lesson and see who was closest.'	Elicitation of Prior Knowledge (Predict-Before-Instruction): By requiring learners to commit to a written prediction before instruction, the teacher surfaces misconceptions (e.g., 'all solids have the same type of bonding'; 'non-metals cannot conduct so they have no free electrons') that will be directly addressed during the Explain phase. Recording predictions on the board creates a public accountability structure that motivates learners to revise their models by the end of the lesson.	Teacher circulates during the 5-minute silent prediction and looks for: (a) learners who leave columns blank — these learners need scaffolding during the Observe phase; (b) learners who correctly sketch electron transfer for NaCl — these can be used as peer-explainers in Phase 3; (c) learners who draw free electrons in aluminium — evidence of prior metallic bonding knowledge. Teacher makes brief notes on a clipboard.
Observe Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12	Part A — Teacher Board Demonstration (12 min): Learners watch and annotate their notebooks as the teacher constructs, step-by-step, dot-and-cross diagrams for: (i) Na → Na^+ and Cl → Cl^- → $[Na^+][Cl^-]$ ionic bond; (ii) H_2 covalent bond (shared pair); (iii) H_2O covalent bonds (two shared pairs, two lone pairs); (iv) O_2 double covalent bond. Learners copy each diagram and, next to each one, write the phrase the teacher uses to name what the	1. Say: 'We are going back to dot-and-cross diagrams you first met in Sub-Strand 2.1. Today we use them to see WHY coastal salt and the sufuria wire behave so differently.' 2. Draw Na electron configuration on the board (2,8,1). Say: 'Sodium has ONE valence electron. It DESPERATELY wants to get rid of it to achieve the noble gas configuration of neon. Watch what happens.' Draw the transfer arrow to Cl (2,8,7). PAUSE 10 SECONDS. Ask: 'What charge	Worked Example with Annotated Think-Aloud: The teacher explicitly narrates the reasoning process ('Sodium DESPERATELY wants to lose…') rather than simply drawing the diagram. This models the scientific practice of Constructing Explanations (NGSS SEP 6) by connecting the observable (electron counts) to the underlying mechanism (bond formation). The pair-practice immediately transfers responsibility to learners, activating the Science and	During pair practice, teacher checks for: (a) correct number of dots and crosses (each element's electrons drawn distinctly); (b) shared pairs positioned between atoms, not on one side; (c) correct total electron count preserved after bonding. Teacher targets at least 4 pairs for a quick 30-second check. Any pair drawing MgO as covalent (a common error) is asked: 'What is the electronegativity difference between Mg and O? Is that a transfer or a share?' — prompting

🔍 Search ARES for similar readings	electrons are doing ('transferred' for ionic; 'shared' for covalent). Part B — Pair Practice (8 min): Each pair receives a card with one assigned molecule from the set: NH_3, Cl_2, HCl, MgO, or CO_2. They draw the dot-and-cross diagram together. One partner draws the diagram; the other writes a caption explaining what happens to electrons. Pairs swap roles halfway through.	does Na now carry? Call out.' [Learners: +1.] 'Why?' WAIT 10 SECONDS. Cold-call one learner by name for the explanation. 3. Complete Na^+Cl^- diagram. Say: 'The electrostatic attraction between Na^+ and Cl^- IS the ionic bond. Write that definition now.' 4. Move to H_2. Say: 'Hydrogen has only one valence electron. It is NOT going to transfer — both atoms want an electron. So they SHARE. This is a covalent bond.' Draw shared pair. 5. Build H_2O and O_2 following the same think-aloud pattern. 6. For pair practice: distribute molecule cards. Say: 'You have 8 minutes. One person draws, one person writes the caption. Swap after 4 minutes. I will cold-call two pairs to present.' Circulate, checking diagrams. If a pair draws NH_3 incorrectly, ask: 'How many valence electrons does nitrogen have? Count them on your diagram.' WAIT 12 SECONDS before offering a hint.	Engineering Practice of Developing and Using Models (NGSS SEP 2).	self-correction before the Explain phase.
Explain Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Six Bond Types Gallery Input (15 min): Teacher introduces the six bond types using a structured notes framework. Learners fill in a partially completed reference table in their notebooks (columns: Bond Type | What holds particles together | Kenyan-context example | Expected melting point: high/low | Conducts electricity? Yes/No/Only when…). The teacher fills in each row live on the board, pausing after each bond type to ask a question. Part B — Bond Type Sorting Activity (10 min): Each pair receives six cards, each describing a substance or material (coastal NaCl, aluminium sufuria, graphite pencil rod, water vapour above a	1. Say: 'Dot-and-cross diagrams show us TWO bond types — ionic and covalent. But there are actually SIX types of chemical bonds. Each one explains a different material in our world. Let us meet all six.' 2. IONIC — Point to coastal NaCl: 'You just drew this one. Electrostatic attraction between oppositely charged ions. Found in our table salt harvested along the Kenya coast. High melting point — it takes 801 °C because the lattice is very strong. Conducts only in solution or when molten — why? Because ions must be FREE to move.' WAIT 10 SECONDS. Cold-call one learner: 'So why does dry salt NOT conduct?' 3. COVALENT —	Partially Completed Reference Table + Physical Card Sort: The partially completed table (NGSS SEP 4 — Analysing and Interpreting Data; SEP 6 — Constructing Explanations) ensures all learners record the key distinctions without copying verbatim, forcing active processing. The card sort operationalises classification (a core science process skill) and requires learners to produce a written justification, making their reasoning visible to the teacher. Connecting each bond type to a named Kenyan material (coastal NaCl, DAP fertiliser, Lake Victoria water, aluminium sufuria, graphite rod) anchors abstract chemistry in	Teacher collects one written card-sort justification sheet from each pair at the end of the activity (or photographs them with a phone). Success indicators: (a) graphite correctly assigned to BOTH covalent (within layers) AND Van der Waals (between layers); (b) NaCl ionic bond correctly distinguished from metallic with reference to ion movement for conductivity; (c) hydrogen bond not confused with a covalent bond. Pairs that misclassify are noted for targeted follow-up in Lesson 5.

	Mombasa beach, iodine crystals used in a school lab test, ammonium ions in DAP fertiliser used in Rift Valley farms). Pairs physically sort the cards under six labelled zones on their desk, then justify each placement in writing using the sentence stem: 'This substance has ___ bonding because ___.'	'Sharing of electron pairs. Found in water, in the O_2 we breathe at Kericho altitude, in the H_2 produced in electrolysis labs. Simple molecular covalent substances have LOW melting points because only weak forces act between molecules — the covalent bonds inside are strong, but the molecules barely attract each other.' 4. DATIVE-COVALENT — 'A special covalent bond where BOTH electrons in the shared pair come from the SAME atom. Found in the ammonium ion, NH_4^+ — important for farmers in the Rift Valley who use DAP fertiliser. Draw the NH_4^+ diagram alongside me.' Walk through the diagram step by step. 5. HYDROGEN BOND — 'A special, relatively strong intermolecular force between molecules where hydrogen is bonded to N, O, or F. This is why water has an unusually high boiling point — very important for life in Lake Victoria. Without hydrogen bonds, Lake Victoria would boil dry.' Ask: 'Is a hydrogen bond the same as a covalent bond? Discuss with your partner for 30 seconds.' Cold-call one pair. 6. METALLIC — Point to aluminium wire: 'A sea of delocalised electrons surrounding a lattice of positive metal ions. The free electrons explain EVERYTHING about the sufuria — conductivity, malleability, shiny appearance, lower melting point than NaCl.' Say: 'Tell me — using what you now know about metallic bonds — why the aluminium BENDS instead of shattering. 15 seconds, think.' Cold-call two learners. 7. VAN DER WAALS — Point to graphite rod: 'The weakest intermolecular force — temporary,	a familiar context and continuously threads back to the anchoring phenomenon.	

		induced dipole attractions between non-polar molecules. Found between iodine molecules, and critically, between the LAYERS in graphite. This is why graphite layers slide off onto paper when you write.' Ask: 'So in graphite, we have TWO bond types at play. What are they, and where?' WAIT 15 SECONDS. Cold-call one learner. 8. For the sorting activity: 'You have the six substance cards. Sort them. Use the sentence stem on your card sheet. You have 10 minutes.' Circulate and specifically target pairs sorting graphite — ask: 'Does graphite have one bond type or two? Look at your notes.' WAIT 12 SECONDS.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes on a sticky note (or a torn piece of paper if sticky notes are unavailable) one NEW question that today's lesson has raised for them — something they still do not understand or want to explore further. They also write one thing they are now MORE confident about explaining. Learners post their questions on the class DQB under whichever kitchen item (salt, sufuria, graphite) the question relates to. The teacher reads three questions aloud, categorises them on the board under the headings 'Bond Formation', 'Structure', and 'Properties', and explicitly flags which upcoming lessons (Lessons 5–8) will answer each question. Learners record in their notebooks: 'By the end of today, I can explain ___ about the anchoring phenomenon, but I still cannot explain ___.'	1. Say: 'We have now named all six bond types. But real understanding is knowing what questions you STILL have. Take 3 minutes — write your new question on your slip of paper, and write one thing you can now explain about our three kitchen items that you could NOT explain at the start of today.' 2. After 3 minutes: 'Come and post your question on the DQB under the item it relates to.' Allow 2 minutes for posting. 3. Read three questions aloud — choose one per kitchen item. Say: 'This is a great question about the salt — specifically about WHY it needs to be molten to conduct. We will answer that in Lesson 6 when we study giant ionic structures.' Repeat for sufuria (metallic structure, Lesson 7) and graphite (giant covalent/layered structure, Lesson 5). 4. Say: 'Every question on that board will be answered before the end of this unit. Your curiosity is driving our learning.' 5.	Driving Question Board as a Living Epistemic Tool: Posting questions publicly transforms individual confusion into collective inquiry goals (NGSS SEP 1 — Asking Questions and Defining Problems). By categorising questions under 'Bond Formation', 'Structure', and 'Properties', the teacher models how scientists organise what they know versus what they still need to investigate. Explicitly mapping DQB questions to future lessons gives learners a coherent narrative arc and prevents the feeling that chemistry is a disconnected set of facts.	The written DQB slip is a direct window into each learner's conceptual state. Teacher scans all slips after the lesson and identifies: (a) learners whose 'I can now explain' statement is vague ('I understand bonds better') — these learners need a more structured graphic organiser in Lesson 5; (b) learners who ask structurally sophisticated questions ('Why does the ionic lattice have a fixed ratio of ions?') — these learners can be given extension reading on crystal lattice structures; (c) whether the majority of questions cluster around one bond type — indicating that that type needs re-teaching.

		Point back to the board predictions from Phase 1: 'Look at your original predictions. How close were you? How has your model changed today?' WAIT 10 SECONDS. Cold-call two learners.		
Model Building Phase 📹 **VIDEO:** [Van der Waals Forces - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the 3-column particle-level sketches they drew in Phase 1. They now REVISE each sketch in a different colour, applying what they have learned about bond types. For NaCl: they replace any vague 'particle' drawings with labelled Na^+ and Cl^- ions held by electrostatic attraction. For aluminium: they draw a lattice of Al^{3+} ions surrounded by a 'sea' of delocalised electrons, labelling it 'metallic bond'. For graphite: they draw hexagonal layers of carbon atoms connected by covalent bonds within each layer, with an arrow and label 'Van der Waals forces' between the layers. Under each revised sketch, learners write one sentence connecting their model to one observable property from the anchoring phenomenon (e.g., 'The delocalised electrons in aluminium can move freely, which is why the sufuria wire conducts electricity as a solid').	1. Say: 'You have 8 minutes to revise your Phase 1 sketches. Use a different colour. Add correct labels. Most importantly — write one sentence under each sketch that connects your model to something we OBSERVED about the three kitchen items at the start of this unit.' 2. Circulate. If a learner draws graphite with only one bond type, ask quietly: 'You have drawn the bonds WITHIN a layer. What about between the layers? What did we say slides off onto paper?' WAIT 12 SECONDS. 3. With 2 minutes remaining, ask the class: 'Hold up your revised aluminium sketch. I want to see the sea of electrons.' Scan the room. Praise specific, accurate drawings: 'I can see Akinyi has drawn the positive ion cores AND the free electrons — that is exactly the metallic bond model.' 4. Close the lesson by saying: 'Today you built the foundation. Every lesson from here will add detail to these models until you can fully answer our driving question: WHY do the salt, the sufuria, and the graphite look the same but behave so differently? Next lesson, we go deep into structure.'	Cumulative Model Revision in Contrasting Colours: Using a second colour to revise the original sketch makes the cognitive change visible — learners can literally see how their model has grown. This practice aligns with NGSS SEP 2 (Developing and Using Models) and gives the teacher a rapid visual assessment of conceptual change. Requiring a written sentence linking the model to an observable property from the phenomenon (rather than just a label) demands that learners bridge micro (particle model) and macro (observable property) levels — the core challenge in chemical bonding education.	Teacher collects (or photographs) three randomly selected notebooks at the end of class. Success criteria for revised models: (a) NaCl — Na^+ and Cl^- labelled with correct charges; electrostatic attraction indicated; (b) Aluminium — positive ion lattice AND delocalised electron sea both present; metallic bond labelled; (c) Graphite — hexagonal covalent layer drawn; Van der Waals forces between layers labelled; at least one layer offset from the next. The connecting sentence must reference a specific observable property (conductivity, melting point, or mechanical behaviour) from the anchoring phenomenon. If the sentence is missing or too vague, the teacher provides written feedback before returning the notebook at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. PREDICTION QUALITY: Were learners' Phase 1 sketches richer or poorer than expected? Did most learners already have a partial correct model for any of the three items? If the aluminium sea-of-electrons model was widespread, consider extending metallic bonding time in Lesson 7 to handle more sophisticated questions.

2. DOT-AND-CROSS STRUGGLES: Which assigned molecule caused the most difficulty in pair practice? If O_2 (double bond) caused widespread confusion, open Lesson 5 with a five-minute revisit before moving to giant structures.

3. SIX BOND TYPES PACING: Was 15 minutes sufficient for all six bond types? If dative-covalent and Van der Waals felt rushed, consider moving Van der Waals to the opening of Lesson 5 (it connects directly to graphite's layered structure) and spending the saved time reinforcing dative-covalent with the NH_4^+ diagram.

4. DQB QUALITY: What proportion of DQB questions were at the 'structure explains property' level versus surface-level ('what is a Van der Waals force?')? A high proportion of surface questions suggests learners need more structured scaffolding in the reference table before the card sort.

5. MODEL REVISION: Were revised sketches genuinely different from Phase 1 sketches, or did learners just re-label without restructuring? If models were superficially revised, begin Lesson 5 with a whole-class model critique activity before introducing new content.

6. EQUITY CHECK: Were the same learners being cold-called repeatedly? Review your cold-call log and ensure gender balance and inclusion of quieter learners in Lessons 5 and 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed (via teacher board demonstration) how valence electrons behave during bond formation: Na transfers its single valence electron to Cl, producing oppositely charged ions held by electrostatic attraction; H, O, and other non-metals share electron pairs to form covalent bonds. They also observed dot-and-cross diagrams for H_2, H_2O, and O_2 and practised drawing diagrams for assigned molecules (NH_3, Cl_2, HCl, MgO, CO_2) in pairs.
What did I learn?	Students learned the definitions and distinguishing features of all six bond types — ionic (electron transfer, e.g., coastal NaCl), covalent (electron sharing, e.g., O_2 and H_2O), dative-covalent (both shared electrons from one atom, e.g., NH_4^+ in DAP fertiliser), hydrogen bonds (H bonded to N/O/F, e.g., water in Lake Victoria), metallic (delocalised electron sea, e.g., aluminium sufuria), and Van der Waals forces (temporary induced dipoles, e.g., graphite layers and iodine crystals). They learned to connect each bond type to expected properties (conductivity, melting point, mechanical behaviour).
How does this explain the phenomenon?	Students explained, through revised particle-level sketches and written sentences, how the bond type inside each kitchen item accounts for one observable property from the anchoring phenomenon: the electrostatic lattice of Na^+Cl^- explains the salt's high melting point and conductivity only in solution; the sea of delocalised electrons in aluminium explains solid-state conductivity and malleability; and the combination of strong covalent bonds within graphite layers and weak Van der Waals forces between layers explains both graphite's electrical conductivity (delocalised electrons within layers) and its ability to flake off onto paper.

LESSON 5 (40 minutes): Practical Investigation of Physical Properties — Connecting Evidence to the Anchoring Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out a structured practical investigation to measure and record the physical properties of four substances — NaCl, candle wax, graphite and aluminium — and use the resulting class data to build evidence that links observable behaviour to bond type and structure.
Knowledge	Learners will be able to state the expected physical properties (solubility in water, electrical conductivity in solid state, relative hardness or brittleness) for substances with giant ionic, simple molecular, giant atomic layered and giant metallic structures, and explain why each property arises from the bonding arrangement holding the particles together.
Skills	Learners will practise safe laboratory technique, systematic data recording in a shared class table, pattern identification across multiple variables, and evidence-based reasoning that connects macroscopic observations to particle-level explanations.
Attitudes	Learners will demonstrate curiosity when unexpected results challenge their predictions, show respect for shared laboratory equipment and for the contributions of every group during whole-class data sharing, and persist in seeking particle-level explanations rather than accepting surface descriptions.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	This lesson is the primary evidence-gathering session in the sequence. Learners move from diagrams and simulations (Lessons 2 to 4) to hands-on measurement, generating first-hand data that will fuel the structured sensemaking discussion in Lesson 6. The four substances tested directly mirror the anchoring phenomenon: NaCl from the Kenyan coast, graphite from a used AA battery, and aluminium from a sufuria, with candle wax added as a contrasting simple molecular substance. Every result recorded today becomes evidence that learners will cite when they write their final explanation in Lesson 8.
Safety Notes	

B. LESSON OVERVIEW
Learners are organised into groups of four. Each group receives a set of four labelled samples (NaCl crystals, a small piece of candle wax, a graphite rod segment, and a strip of aluminium foil) and a simple conductivity tester (battery, two crocodile-clip leads, and an LED or small bulb). Groups test each sample for three properties: (1) solubility — does a small amount dissolve in 5 mL of water in a plastic cup? (2) electrical conductivity — does the LED light when the sample completes the circuit? (3) mechanical behaviour — is the sample hard and brittle, soft, or flexible/malleable? Results are recorded on a structured group worksheet. After 20 minutes of practical work the teacher calls time and leads a 10-minute whole-class data-sharing session, building a single master table on the board. A 5-minute cold-call discussion asks learners to identify the most surprising result and to connect it back to the anchoring phenomenon. The final 5 minutes are used to update the Driving Question Board and set up the sensemaking challenge for Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Before touching any equipment,	Teacher circulates during the 2-	Prediction table activates prior	Teacher scans prediction tables

📹 **VIDEO:** [Precipitation reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Power and Electrical Energy Use (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	learners open their notebooks to a fresh page and draw a prediction table with rows for NaCl, candle wax, graphite and aluminium, and columns for solubility, conductivity (solid) and mechanical behaviour. Working individually for 2 minutes, learners write their predictions using the bond-type knowledge from Lessons 2 to 4 — for example a learner might write: NaCl — does not conduct as solid because ions are locked in place, candle wax — does not conduct because it has no ions or free electrons, graphite — should conduct because delocalised electrons exist between layers, aluminium — conducts because of the delocalised electron sea. Learners then compare predictions with their group partner for 1 minute before the practical begins, noting any disagreements on a sticky note.	minute individual writing time and reads predictions over shoulders without commenting, noting which misconceptions are live — for example learners who predict that graphite will NOT conduct because it is a non-metal. Teacher says: 'Do not change your predictions yet — you will need them to evaluate your results later. Write what you actually believe right now, based on what we learned in Lessons 2 to 4.' Teacher then poses the pre-lab prompt aloud: 'Think about the structure of each substance at the particle level. What is moving — or NOT moving — inside each solid that would allow or prevent electricity to flow?' WAIT TIME: 30 seconds of silence before accepting any verbal response. Cold-call one learner to share their graphite prediction, specifically asking: 'What did we learn about graphite structure in Lesson 4 that is relevant here?'	knowledge from Lessons 2 to 4 and creates cognitive dissonance when graphite (a non-metal) is predicted by some learners to be non-conducting, setting up a productive surprise during the Observe phase.	during circulation — if more than half the class predicts graphite is non-conducting, teacher makes a mental note to pause on that data point during whole-class sharing. No correction is given at this stage; predictions are intentionally preserved as evidence to revisit.
Observe Phase 📹 **VIDEO:** [Precipitation reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Power and Electrical Energy Use (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Groups carry out the three tests in sequence over 20 minutes. Test 1 — Solubility: Each group adds a small spatula tip (approximately 0.5 g) of each substance to separate labelled plastic cups containing 5 mL of water, stirs for 30 seconds with a glass rod and records whether the substance dissolves (clear solution), partially dissolves or remains unchanged. For candle wax learners observe that shavings float and do not dissolve. For NaCl learners observe a clear solution forming. For graphite and aluminium learners observe that particles remain. Test 2 — Electrical conductivity: Learners connect each sample in turn between the two crocodile clips of the	Teacher circulates continuously, group by group, asking probing questions rather than giving answers. Suggested moves: At the conductivity station — 'Your LED just lit up for graphite. Does that surprise you? Write down why or why not before moving on.' At the NaCl station — 'The dry salt does not light the bulb, but we saw in Lesson 1 that salt solution does light a bulb. What is different between solid salt and dissolved salt at the particle level?' WAIT TIME: 20 seconds after each probe before accepting a group response. At the candle wax station — 'What type of structure does candle wax have? What forces hold the molecules together? How would those forces	The practical is structured so that at least two results are likely to contradict common prior-knowledge misconceptions: graphite conducting (surprises learners who believe only metals conduct) and dry NaCl not conducting (surprises learners who associate salt with electricity from the Lesson 1 phenomenon). These surprises are intentional cognitive anchors for the Explain phase.	Teacher notes which groups are struggling to make the LED light with graphite (poor contact — coach them to press clips firmly on opposite ends of the rod, not the same face). Teacher listens for correct use of vocabulary: delocalised electrons, ions, intermolecular forces. Any group that explains the graphite result correctly using delocalised electrons is flagged to share during the cold-call segment.

	conductivity tester (battery plus LED circuit). They press the clips firmly onto opposite faces of each sample, observe whether the LED glows (conducts) or remains dark (does not conduct), and record the result. The graphite result — LED glows — is expected to surprise learners who predicted non-conduction. The NaCl dry crystal result — LED does not glow — is expected to surprise learners who associate salt with conductivity from Lesson 1. Test 3 — Mechanical behaviour: Learners attempt to (a) bend the aluminium foil strip, (b) press a finger firmly onto the NaCl crystal edge, (c) snap the graphite rod segment gently, and (d) scratch the candle wax surface with a fingernail. They record descriptors: malleable and flexible (aluminium), hard and slightly brittle at edges (NaCl), brittle and snaps (graphite), soft and easily scratched (candle wax). Results are recorded in the structured group worksheet using the class data table format.	explain what you are seeing?' Teacher does NOT resolve contradictions during the practical — contradictions are preserved for the whole-class sharing discussion. If a group finishes early, teacher directs them to re-examine their prediction table and annotate any predictions that their data has already challenged, using a different coloured pen.		
Explain Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After practical work ends (20-minute mark), the teacher projects or draws the master class data table on the board with columns: Substance, Solubility in water, Conducts as solid (yes or no), Mechanical behaviour, Likely structure type. Teacher calls on one reporter per group (rotating so each lesson a different learner reports) to give their group result for one substance. Once the table is complete, learners spend 3 minutes individually annotating their own worksheet: next to each result they write a one-sentence particle-level explanation connecting the observation to	Teacher builds the master table on the board by cold-calling reporters. After the table is complete, teacher says: 'You now have experimental evidence in front of you. I want you to look at the entire table for 30 seconds in silence and identify the single result that surprises you most.' WAIT TIME: 30 seconds of strict silence — teacher counts internally and does not speak. Then cold-calls three individual learners (not group reporters — choose learners who have been quiet so far): Cold-call 1: 'Tell me the result that surprised you and explain in ONE sentence what it tells us about the particles inside	The master class table aggregates all groups data into a single evidence base, reducing noise from individual experimental error and allowing pattern recognition across four substances. The 30-second silence before cold-call is a structured wait-time strategy that prevents fast responders from dominating and gives all learners processing time. Connecting each result to the anchoring phenomenon items re-anchors the lesson in the driving question.	Teacher listens to the three cold-call responses and uses them diagnostically: if learners cannot connect the data to particle-level explanations, teacher notes this as a priority for the Lesson 6 sensemaking session. If learners ARE using vocabulary like delocalised electrons and lattice correctly, teacher explicitly affirms the language: 'That is exactly the vocabulary a scientist would use — hold onto that phrase for Lesson 6.'

	bond type. For example: 'Graphite conducts because delocalised electrons between the carbon layers are free to move and carry charge' and 'Dry NaCl does not conduct because the Na+ and Cl- ions are locked in fixed lattice positions and cannot move to carry charge — they only become mobile when dissolved or melted.' The teacher then facilitates a 5-minute structured discussion using three cold-call questions (see Teacher Moves). Learners connect each result explicitly back to the anchoring phenomenon items: aluminium sufuria (metallic bonding), graphite from the battery (layered giant atomic), NaCl from the coast (giant ionic), candle wax (simple molecular as a new contrast substance).	that substance.' Cold-call 2: 'Connect the result you just heard to one of the three items from our demonstration at the start of this unit — the salt, the aluminium wire or the graphite rod.' Cold-call 3: 'Our driving question asks why these three kitchen items behave so differently. Based on today's data alone, what is your best answer right now — even if it is incomplete?' After the third cold-call teacher summarises: 'Today we generated evidence. In Lesson 6 we will use this evidence to build a complete explanation. Your job now is to make sure your data table is accurate and your annotations are in your own words — because you will need them next lesson.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, learners return to the class Driving Question Board displayed on the wall or board. Each group has two sticky notes (different colours if available — one colour for answers, one for new questions). On the ANSWER note, groups write one claim supported by their practical evidence — for example: 'We now have evidence that graphite conducts electricity even though it is a non-metal, which means conductivity depends on whether electrons can move freely, not on whether the substance is a metal.' On the QUESTION note, groups write one new question raised by the data — for example: 'If graphite conducts, why does diamond (also pure carbon) not conduct?' or 'Why does NaCl conduct when dissolved but not when solid — what exactly changes when it dissolves?'	Teacher circulates as groups write their sticky notes, coaching learners to make claims evidence-based: 'Do not write what you believe — write what the data shows. Start your claim with: Our data shows that...' Teacher selects the three most generative new questions from the DQB to read aloud, choosing ones that directly set up the Lesson 6 sensemaking discussion (e.g., the diamond vs graphite question, the solid vs dissolved NaCl question, and a question about why candle wax has such a low melting point). Teacher closes by pointing to the driving question posted prominently: 'Every question on this board is a piece of the puzzle. The puzzle is: what is happening between the atoms that controls every property we can measure? You now have real experimental data to work with. Next lesson we	The two-colour sticky-note protocol separates evidence-based claims from new questions, making the epistemic status of each contribution explicit. Reading three questions aloud models the scientific practice of using evidence to generate further inquiry rather than treating data as a final answer.	Teacher scans all answer sticky notes before the next lesson. Any note that states a claim without evidence (e.g., 'Salt does not conduct because it is ionic' with no reference to the data) is flagged for follow-up at the start of Lesson 6, prompting the learner to revise the claim to include the observed evidence. Notes that correctly link observation to particle-level explanation are used as model answers in Lesson 6.

	Groups post both notes on the DQB. Teacher reads three question notes aloud and says: 'These questions are exactly what Lesson 6 is designed to answer. Come ready with your data table and annotations.'	explain it.'		
Model Building Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Power and Electrical Energy Use (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	This phase is embedded as a brief 3-minute notebook activity rather than a full model-building session (full model building occurs in Lesson 8). Each learner selects ONE substance from the practical and draws a quick particle diagram in their notebook — not a formal dot-and-cross diagram but a simplified sketch showing the key structural feature that explains their most surprising experimental result. For example: a learner choosing graphite draws the layered hexagonal structure with arrows indicating delocalised electrons moving between layers and labels it 'free electrons carry charge — LED lights up.' A learner choosing NaCl draws a section of the ionic lattice with Na+ and Cl- locked in place and labels it 'ions cannot move in solid — LED stays dark.' The sketch is annotated with the specific experimental observation it explains. This serves as a personal model revision in response to today's evidence.	Teacher gives a clear instruction: 'In the next 3 minutes, draw one particle sketch that explains the result that surprised you most today. Label it with the observation from your data table. This is not assessed right now — it is your thinking on paper.' Teacher circulates and looks for diagrams that correctly represent the structural feature responsible for the property. For graphite sketches teacher checks that the learner has indicated electron mobility between layers, not just drawn hexagons. For NaCl sketches teacher checks that the learner has drawn ions in fixed positions. Teacher says quietly to any learner drawing an incorrect model: 'Look back at the Lesson 4 structure diagram in your notebook — what do those layers in graphite have that the NaCl lattice does not have?' WAIT TIME: allow 10 seconds for the learner to think before adding any further hint.	The quick particle sketch forces learners to translate tabular data into a spatial, particle-level representation — a core scientific modelling practice. It also serves as a low-stakes model revision moment: learners who held the misconception that graphite cannot conduct must now reconcile their sketch with the evidence they generated, creating the cognitive need that Lesson 6 will resolve.	Teacher photographs three or four representative notebook sketches (with learner permission) using a phone or tablet to project at the start of Lesson 6 as discussion anchors, showing a range from partially correct to fully correct models. This gives the Lesson 6 discussion a concrete starting point grounded in learner thinking rather than a textbook diagram.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Which of the four substances produced the most disagreement or surprise in learner predictions — and does that match your expectation that graphite and dry NaCl would be the two most challenging results? (2) Were learners able to move from describing observations (the LED lit up) to explaining observations (the LED lit up because delocalised electrons in graphite are free to move and carry charge)? If not, what additional scaffolding do you need to build into the Lesson 6 sensemaking discussion? (3) Did all groups generate usable data, or did equipment issues (poor LED contact, insufficient sample size) compromise results for some groups? If so, plan to present the full class data set at the start of Lesson 6 rather than relying on individual group data. (4) Review the DQB question notes: do the new questions cluster around diamond vs graphite, or around the solid vs dissolved NaCl distinction? The dominant cluster should shape

how much time you allocate to each explanation in Lesson 6. (5) Were the Kenyan-context connections (coastal NaCl, sufuria aluminium, battery graphite) genuinely motivating for learners, or did they feel superficial? Consider asking learners directly at the start of Lesson 6: 'Which of the four substances we tested today do you encounter most often at home — and does knowing its structure change how you think about using it?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that: NaCl crystals do not conduct electricity as a solid but dissolve in water; candle wax does not dissolve in water and does not conduct; graphite conducts electricity as a solid despite being a non-metal; aluminium conducts electricity as a solid and is flexible and malleable rather than brittle. Learners also observed that candle wax is soft and easily scratched, NaCl is hard, graphite is brittle and snaps, and aluminium bends without breaking.
What did I learn?	Learners learned that the physical properties of a solid substance — its solubility, electrical conductivity and mechanical behaviour — are directly determined by the type of bonding and structure holding its particles together. They learned that conductivity in a solid requires particles (electrons or ions) that are free to move, and that this freedom depends entirely on bond type: metallic and layered giant atomic structures allow electron movement, while giant ionic and simple molecular structures do not (in the solid state).
How does this explain the phenomenon?	Learners can now explain the anchoring phenomenon more completely than in Lesson 1: aluminium conducts and bends because delocalised electrons move freely through the metallic lattice and the layers of metal ions can slide past each other; graphite conducts because delocalised electrons exist between the carbon layers and are free to carry charge, even though carbon is a non-metal; dry NaCl does not conduct because its ions are locked in a rigid lattice and cannot carry charge until the lattice is broken by dissolving or melting; candle wax neither conducts nor dissolves because its molecules are held together only by weak Van der Waals forces and contain no ions or free electrons. The driving question has advanced: learners now have experimental evidence, not just theoretical predictions, to support bond-type explanations of physical properties.

LESSON 6 (80 minutes): Seeing the Invisible: Drawing Ionic and Covalent Bonds with Dot-and-Cross Diagrams	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners navigate online virtual laboratory simulations (or a teacher-led projected simulation) to observe ionic and covalent bond formation at the particle level, then construct and interpret dot-and-cross diagrams for NaCl, MgO, HCl, NH_3 and CH_4 — directly linking electron behaviour to the observable properties of the three kitchen items (salt, aluminium sufuria, graphite rod) that anchor this unit.
Knowledge	Learners will be able to: (a) illustrate chemical bonding in common compounds using dot-and-cross diagrams; (b) identify bond types (ionic vs. covalent) in NaCl, MgO, HCl, NH_3 and CH_4; (c) distinguish electron transfer (ionic) from electron sharing (covalent) at the particle level; (d) describe the relationship between bond type and selected physical properties (conductivity, melting point, brittleness).
Skills	Navigating a virtual laboratory simulation; drawing accurate dot-and-cross diagrams; identifying patterns across multiple representations; peer comparison and error correction; presenting and justifying scientific diagrams to the class.
Attitudes	Curiosity about the invisible particle world; persistence in perfecting diagram accuracy; respect for peers' contributions during cold-call presentations; appreciation that abstract bonding models explain real, observable kitchen-table phenomena.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	Lessons 1–5 established that the three kitchen items (salt, aluminium wire, graphite rod) differ dramatically in conductivity, melting point and mechanical behaviour, and introduced the idea that electrons are unevenly distributed in atoms. Lesson 6 is the critical visualisation lesson: learners see — at the particle level — exactly what electron transfer and electron sharing look like, giving them the mechanistic language needed to explain every anomaly on the DQB. This prepares them for Lesson 7 (giant structures and metallic bonding) and Lesson 8 (full model synthesis).
Safety Notes	Virtual simulation work: ensure all devices are charged or plugged in; school Wi-Fi passwords shared in advance. If using the projected simulation only, learners must not crowd the demonstration bench. Pencils and rulers are the only physical tools; no chemicals are handled in this lesson. Remind learners to sit correctly at screens to avoid eye strain.

B. LESSON OVERVIEW
Learners open the PhET 'Build a Molecule' simulation or an equivalent KICD-approved virtual lab (teacher projects for device-limited settings). In the first half of the lesson they observe animated ionic bond formation (NaCl, MgO) and covalent bond formation (HCl, NH_3, CH_4) at the electron level. In the second half they work in pairs to complete a structured dot-and-cross worksheet for all five compounds, compare diagrams with another pair, and identify the core pattern: ionic bonds involve complete electron transfer producing charged ions, while covalent bonds involve shared electron pairs keeping atoms neutral. Four pairs are cold-called to draw and explain diagrams on the board. The teacher facilitates whole-class correction. All findings are added to the Driving Question Board, visibly advancing the class's ability to answer why table salt (ionic) behaves so differently from the graphite rod and aluminium sufuria.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners examine a projected	Display the split-image and say:	Elicit-before-instruction prediction	Teacher scans sketches during the

📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparison of Dot, Two-Sided Stem, and Parallel Box Plots (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	split-image showing: (left) a grain of table salt from the Kenyan coast labelled 'NaCl — ionic'; (right) a molecule of water labelled 'H_2O — covalent'. Each learner independently writes answers to two questions on a sticky note or in their exercise book: (1) 'In NaCl, do you think sodium GIVES electrons to chlorine, or do they SHARE electrons? Sketch what you think it looks like.' (2) 'In H_2O, do you think the atoms give or share electrons? Sketch your idea.' Learners then turn-and-talk with their partner for 2 minutes to compare predictions before the teacher takes responses.	'Before we use any simulation today, I want to see what you already think is happening inside these two substances — the salt on your kitchen table and the water you boil ugali in.' Allow 4 minutes of silent individual writing. After individual time say: 'Now compare your sketch with your partner. You have 2 minutes — go.' After partner talk, cold-call THREE learners (choose one who predicted transfer, one who predicted sharing, one who is uncertain): 'Zawadi, tell me your sketch for NaCl — transfer or sharing?' WAIT 12 seconds between each cold-call. Write the three competing ideas on the board under the heading 'OUR PREDICTIONS — Lesson 6'. Do NOT confirm or correct yet. Say: 'We are going to use a simulation today that will show us which of these ideas is right — and maybe surprise some of us.'	(NGSS SEP-2: Developing and Using Models). By externalising prior mental models as sketches before instruction, learners create a cognitive hook that the simulation observation will either confirm or challenge. Writing predictions on the board maintains accountability and gives the teacher a diagnostic snapshot of the class's starting point.	4-minute silent period and identifies: (a) learners who correctly anticipate electron transfer for NaCl (evidence of prior learning); (b) learners who draw sharing for both (common misconception to address); (c) learners who leave the ionic sketch blank (may need scaffold during simulation). Teacher uses this scan to decide which three predictions to display and to position targeted questions during Phase 2.
Observe Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparison of Dot, Two-Sided Stem, and Parallel Box Plots (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners (individually or in pairs sharing a device) navigate to the virtual laboratory simulation. The teacher has pre-loaded two simulation tasks: TASK A — Ionic bond formation: observe Na donating one electron to Cl to form Na^+ and Cl^- (NaCl), then Mg donating two electrons to O to form Mg^{2+} and O^{2-} (MgO). TASK B — Covalent bond formation: observe H and Cl sharing one electron pair (HCl), N and three H atoms sharing three pairs (NH_3), and C sharing four pairs with four H atoms (CH_4). For each compound learners complete an observation table in their worksheets with columns: Compound | Atoms involved | Electrons transferred OR shared |	Before releasing learners to devices, model the simulation navigation for 2 minutes: 'I am going to show you NaCl. Watch what happens to the electron — I will pause it here. What did you see? Call it out.' Restart and pause again at the moment of electron transfer. After the model demo say: 'You have 20 minutes. Complete every row in your observation table. If you finish early, re-run the NH_3 simulation and count the shared pairs.' Circulate every 3–4 minutes. Stop the class at the 10-minute mark for a mid-observation check: cold-call TWO pairs — 'Baraka and Amina, what charge does sodium carry after losing one electron — and why?' WAIT 12 seconds. 'Good —	Structured observation with a comparison table (NGSS SEP-3: Planning and Carrying Out Investigations; SEP-4: Analysing and Interpreting Data). The table forces learners to notice the contrast between Tasks A and B — ionic bonding produces charged ions (explaining why dissolved NaCl conducts electricity) while covalent bonding produces neutral molecules (explaining why HCl gas does not conduct). The mid-observation cold-calls prevent passive screen-watching and require learners to articulate the mechanism in their own words.	Teacher checks completion of observation tables during circulation. Specific indicators: (a) learner correctly writes Na^+ and Cl^- (not Na and Cl) after ionic bond formation; (b) learner correctly writes '4 shared pairs' for CH_4; (c) learner correctly writes 'neutral' in the charge column for covalent molecules. Any learner writing a charge for a covalent molecule receives immediate individual redirection: 'Look at your simulation again — does the H atom in HCl carry a + or − sign after bonding?'

	Charge on each particle after bonding | Bond type. In device-limited settings, the teacher drives the simulation on the projector and pauses at each compound for learners to complete the table row before moving on.	now Peter and Faith, how is MgO different from NaCl in terms of the number of electrons moved?' WAIT 12 seconds. At the 18-minute mark, pause the class for covalent bonds: 'Before you start Task B, predict with your partner: will the atoms gain charges like in Task A?' Allow 60 seconds of partner prediction, then release. At the 26-minute mark cold-call ONE pair for CH_4: 'How many shared pairs does carbon form — and connect that to carbon's position in the periodic table.' WAIT 15 seconds.		
Explain Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparison of Dot, Two-Sided Stem, and Parallel Box Plots (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Pairs complete a structured dot-and-cross worksheet with five tasks, drawing full diagrams for: (1) NaCl — show outer shell electrons of Na (2,8,1) and Cl (2,8,7), electron transfer, and the resulting ions in square brackets with charges; (2) MgO — show Mg (2,8,2) and O (2,6), double transfer; (3) HCl — show H (1) and Cl (2,8,7), one shared pair; (4) NH_3 — show N (2,5) and three H atoms, three shared pairs and one lone pair; (5) CH_4 — show C (2,4) and four H atoms, four shared pairs. After completing their own diagrams, each pair swaps worksheets with the pair behind them, compares diagrams, and circles any differences in green pencil with the comment 'CHECK: transfer or share?' The teacher then cold-calls FOUR pairs to draw their diagrams on the board — one pair per compound (NaCl, HCl, NH_3, CH_4), with MgO drawn by the teacher as a worked example first. After each board diagram, the teacher facilitates a 90-second whole-class correction round.	Distribute the printed worksheet (or display it on screen). Say: 'Use dots for one atom's electrons and crosses for the other atom's electrons — that way we can track exactly where each electron came from. You have 15 minutes to complete all five diagrams.' Circulate; when you see a learner drawing a full electron shell for H (i.e., writing 2,8 for hydrogen), stop and say quietly: 'Hydrogen only has one shell — how many electrons does it need to feel stable?' WAIT 10 seconds. After the 15-minute pair-work period say: 'Swap your worksheet with the pair behind you. You have 4 minutes to compare and mark any differences. Use your green pencil.' After peer-comparison, announce the four cold-called pairs by name: 'Fatuma and Korir — NaCl on the board, please. Achieng and Mutua — HCl. Njeri and Odhiambo — NH_3. Wanjiku and Kipchoge — CH_4.' While pairs draw, circulate and coach quietly. After each diagram is on the board: 'Class — what is correct here? What needs to change? Tell	Representation construction and peer critique (NGSS SEP-2: Developing and Using Models; SEP-6: Constructing Explanations). Drawing dot-and-cross diagrams is not rote copying — it requires learners to apply electron configuration knowledge (from Sub-Strand 2.1) to predict which electrons move or are shared. The peer-swap step activates collaborative error-detection, and the public board presentations with whole-class correction create a shared, authoritative reference that the class can return to in later lessons. The synthesis question explicitly connects the diagram pattern to the anchoring phenomenon: ionic bonding produces ions that carry charge (explaining NaCl conductivity in solution) while covalent bonding produces neutral molecules.	Observable indicators during worksheet circulation: (a) learner correctly places electron-transfer arrow in NaCl diagram; (b) learner draws square brackets and correct charges $[Na]^+$ and $[Cl]^-$; (c) learner draws a bonding pair as two dots/crosses between nuclei in HCl (not on only one atom); (d) learner shows the lone pair on N in NH_3. Teacher notes the names of any learners still struggling with lone pairs for targeted support in Lesson 7. Board presentations assessed against a displayed answer key revealed only after all four pairs have drawn.

		your partner first.' WAIT 10 seconds, then cold-call one observer: 'Otieno, one thing that needs fixing on the NaCl diagram?' Correct on the board together. After all four diagrams are corrected, ask the synthesis question: 'Look at the NaCl diagram and the CH_4 diagram side by side. In one sentence — what is the biggest difference in what happens to the electrons?' WAIT 15 seconds, then cold-call TWO learners.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparison of Dot, Two-Sided Stem, and Parallel Box Plots (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky-note contributions to the DQB: one CLAIM (something they now understand) and one REMAINING QUESTION (something the diagrams have raised). Teacher reads selected contributions aloud and physically moves sticky notes to new columns on the DQB: 'Evidence we now have', 'New questions the evidence raises', and 'Still unexplained'. The class discusses which of the original three kitchen-item mysteries (salt conductivity, aluminium conductivity, graphite conductivity) can now be partly explained using today's dot-and-cross evidence, and which still require further investigation.	Hand out two sticky notes per learner (different colours for CLAIM vs. QUESTION). Say: 'You have 3 minutes — one sticky note says something you can now explain about our three kitchen items. The other sticky note asks a question today's diagrams raised but did not answer.' Collect and read FOUR claim notes aloud, choosing notes that specifically mention the phenomenon: e.g. 'NaCl has ions because Na transfers its electron to Cl — that is why it conducts when dissolved.' After reading, say: 'Can we now explain why dry salt does not conduct electricity? Turn and tell your partner — you have 45 seconds.' Cold-call ONE pair. Then display the DQB and say: 'We understand ionic and covalent bonds at the particle level now. But our aluminium sufuria conducts electricity as a solid — and aluminium forms metallic bonds, not ionic or simple covalent ones. And our graphite conducts too. What is happening in those structures?' Move the 'aluminium' and 'graphite' sticky notes to 'Still unexplained — needs Lesson 7.' Say: 'Write that question on your	Claim-Evidence-Reasoning (CER) sticky-note protocol. By requiring learners to write a CLAIM grounded in today's simulation and diagram evidence, the DQB update transforms individual lesson activities into a cumulative, publicly visible argument. Moving sticky notes to 'Still unexplained' explicitly frames metallic bonding and giant covalent structures (graphite) as the next investigative target, maintaining the story arc of the unit and sustaining intellectual motivation.	Teacher reads all CLAIM sticky notes before the lesson ends to check for two misconceptions: (a) claims that say 'covalent bonds conduct electricity' — redirect to: 'Does CH_4 (methane, used in cooking gas) conduct electricity? Why not?'; (b) claims that conflate ionic bonding in solid NaCl with conduction — redirect to: 'If ions are locked in a lattice, can they move to carry charge?' These misconceptions, if present, are addressed as a whole-class correction in the first 3 minutes of Lesson 7.

		remaining-question sticky note if it matches yours.'		
Model Building Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparison of Dot, Two-Sided Stem, and Parallel Box Plots (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner retrieves their cumulative particle model diagram (started in Lesson 2 and revised in each subsequent lesson). Today they add a new annotated panel for NaCl (ionic) and for CH_4 (covalent), pasting in or redrawing a miniaturised dot-and-cross diagram for each, and writing a one-sentence caption connecting the bond type to one observable property from the anchoring phenomenon. Learners also add a 'mystery panel' with a question mark for aluminium and graphite, labelled 'Metallic bond? Giant covalent? — investigate Lesson 7.'	Say: 'Open your model booklet to your cumulative diagram. Add two new panels — one for NaCl showing its ionic dot-and-cross, and one for CH_4 as a simple covalent example. Under each panel, write one sentence connecting the bond to a property we observed on Day 1 with our three kitchen items.' Allow 6 minutes for independent model revision. With 2 minutes remaining, cold-call TWO learners to read their NaCl caption aloud: 'Kamau, read your NaCl caption — how does ionic bonding connect to the behaviour of salt we saw in the conductivity test?' WAIT 12 seconds. Close the lesson by pointing to the 'mystery panel': 'Aluminium and graphite are still on our board. Next lesson we build the model that explains them — and you will finally be able to answer our driving question completely.'	Cumulative model revision (NGSS SEP-2: Developing and Using Models). The model booklet is the physical artefact of the entire unit's developing understanding. Adding ionic and covalent panels today — while leaving the aluminium and graphite panels blank — makes the incompleteness of the explanation tangible and motivating. The one-sentence caption requirement forces learners to connect abstract particle diagrams to the macroscopic phenomenon observed in Lesson 1, reinforcing the core NGSS practice of linking multiple levels of representation.	Teacher scans model booklets during the 6-minute work period. Specific indicator: learner's NaCl caption explicitly mentions IONS and CHARGE, not just 'electrons move'. Learner's CH_4 caption explicitly mentions SHARING and NEUTRAL. Any learner writing 'NaCl shares electrons' in their caption is flagged for a one-to-one check at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, consider: (1) During the simulation observation (Phase 2), did ALL learners correctly populate the 'charge' column for both ionic and covalent compounds, or did a significant number write charges for covalent molecules? If so, begin Lesson 7 with a 3-minute whole-class correction before introducing metallic bonding. (2) During the board presentations (Phase 3), which of the five compounds produced the most errors — particularly, did learners struggle with the lone pair on N in NH_3 or with double-transfer in MgO? If MgO caused confusion, prepare an additional worked example of CaO at the start of Lesson 7. (3) Review the DQB CLAIM sticky notes: do learners' claims specifically connect ionic bonding to the dry-salt-does-not-conduct observation from the anchoring phenomenon? If claims are too abstract ('ionic bonds involve electron transfer') without connecting to the phenomenon, devote 5 minutes at the start of Lesson 7 to explicitly bridging from dot-and-cross diagrams to the macroscopic conductivity test. (4) Note names of any learners who could not complete the dot-and-cross worksheet in time — consider pairing them with a peer mentor in Lesson 7's group work. (5) Reflect on device availability: if the projected simulation was used rather than individual devices, plan to give learners individual simulation access during the Lesson 7 station rotation so they can self-pace through the metallic bonding animation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using a virtual laboratory simulation (projected or on individual devices), learners observed animated ionic bond formation in NaCl and MgO — watching sodium and magnesium transfer electrons to chlorine and oxygen respectively — and covalent bond formation in HCl, NH_3 and CH_4 — watching atoms share electron pairs without transferring charge.
What did I learn?	Ionic bonding involves complete electron transfer from a metal to a non-metal, producing oppositely charged ions (e.g., Na^+ and Cl^- in table salt from the Kenyan coast); covalent bonding involves electron sharing between non-metals, producing neutral molecules (e.g., CH_4 in cooking gas, NH_3). Learners can now draw accurate dot-and-cross diagrams for all five compounds and explain why ionic compounds produce mobile ions when dissolved (connecting to the salt conductivity observation in the anchoring phenomenon) while simple covalent molecules do not.
How does this explain the phenomenon?	The dot-and-cross diagrams explain why dry table salt does not conduct electricity (ions are locked in a rigid lattice — no mobile charge carriers) but conducts when dissolved in water (ions become free to move). They also explain why CH_4 and similar simple covalent molecules have low melting points (only weak intermolecular forces — explored further in Lesson 7). The aluminium sufuria and graphite rod remain in the 'still unexplained' column of the DQB, motivating Lesson 7's investigation of metallic and giant covalent structures.

LESSON 7 (80 minutes): Bond Types, Structures and Physical Properties: Making Sense of the Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise evidence gathered across Lessons 1–6 to construct explicit explanations for why the aluminium sufuria, graphite pencil rod and table salt behave differently, by mapping each substance's bond type and particle-level structure directly onto its measured physical properties.
Knowledge	Learners identify and describe the four structural types — giant ionic (NaCl), giant metallic (Al), giant atomic/covalent (graphite/diamond) and simple molecular — and explain how the arrangement and mobility of particles and electrons within each structure determines melting point, electrical conductivity, solubility and mechanical behaviour.
Skills	Learners construct written cause-and-effect explanations using evidence from prior lessons; create annotated structure diagrams that link microscopic features to macroscopic properties; and evaluate claims made by peers using the CER (Claim–Evidence–Reasoning) framework.
Attitudes	Learners appreciate that selecting appropriate materials for everyday use (sufuria, pencil, seasoning) requires understanding the particle-level structure of those materials, and reflect on the responsibility of informed consumer and engineering choices.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	Lessons 1–6 gathered raw evidence: conductivity tests, melting-point data, solubility observations, dot-and-cross diagrams, and virtual-lab simulations. Lesson 7 is the Explain phase — learners now consolidate all evidence into coherent, structured explanations that directly answer the driving question. By the end of this lesson, each structural type is explicitly mapped to its property profile, and learners can predict the properties of an unfamiliar substance from its bond type alone.
Safety Notes	No hazardous chemicals are used in this lesson. If any physical samples (salt crystals, aluminium wire, graphite rod) are handled as reference objects during discussion, learners should wash hands afterwards. Ensure electrical circuit demonstration equipment (batteries, bulbs, wires) is handled with dry hands only. Broken graphite rods have sharp edges — handle with care and dispose of fragments in a solid-waste bin.

B. LESSON OVERVIEW

This 80-minute Explain-phase lesson is the seventh in the eight-lesson Chemical Bonding sequence. Learners work in groups of four to construct written CER (Claim–Evidence–Reasoning) explanations for each of the three anchoring-phenomenon substances — table salt (NaCl), aluminium (Al) and graphite (C) — drawing exclusively on data and observations recorded in their science notebooks during Lessons 1–6. Groups then share explanations in a structured gallery walk, critique each other's reasoning, and co-build a whole-class consensus explanation wall. The teacher uses Socratic questioning, targeted cold-calling and a 'Press for Mechanism' move to push explanations from surface description ('salt doesn't conduct because it is ionic') to particle-level mechanism ('in solid NaCl the Na^+ and Cl^- ions are locked in fixed lattice positions and cannot move to carry charge; dissolving in water or melting frees them to move'). The lesson closes with a predictive transfer task: learners apply their structural framework to two new Kenyan-context substances — iodine crystals used in water purification tablets sold in Nairobi pharmacies, and the iron in a Jua Kali welding rod — and predict their properties before the teacher confirms. The DQB is updated and the cumulative particle model is revised.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their science notebook and, working individually for 5 minutes, writes responses to the PREDICT prompt displayed on the board: 'Without looking anything up, write one sentence explaining WHY dry table salt does not conduct electricity, one sentence explaining WHY aluminium conducts even as a solid, and one sentence explaining WHY graphite conducts even though carbon is a non-metal. Use the words PARTICLES, ELECTRONS and STRUCTURE in each sentence.' Learners then share with a shoulder partner (Think-Pair-Share, 2 minutes). Pairs identify where their explanations agree or differ and flag one unresolved disagreement to share with the class.	Display the PREDICT prompt on the board before learners enter the room. Say: 'This is not a test — I want to see your thinking RIGHT NOW, before we dig deeper. Write your best explanation even if you are not sure.' Allow 5 minutes of individual silent writing — do not intervene. After Think-Pair-Share, cold-call FOUR pairs (select strategically: one pair with a strong mechanistic answer, one with a surface-level answer, one with a misconception, one with an honest 'I am not sure'). For each pair say: 'Tell us your disagreement.' After each response, apply 10-second WAIT TIME before commenting. Do NOT correct errors yet — instead say: 'Hold that thought — let's see if our evidence from the past six lessons can settle this.' Record 3–4 student prediction phrases verbatim on the board under the heading 'Our predictions before sensemaking'.	Think-Pair-Share with Disagreement Surfacing. By asking pairs to flag disagreements rather than agreements, the teacher surfaces productive misconceptions (e.g. 'graphite conducts because it is shiny like a metal') that will be resolved during the Explain phase. Recording predictions verbatim creates accountability — learners will return to these phrases at the end of the lesson to self-assess how their thinking has changed.	Circulate during individual writing (3 minutes). Look for: (a) whether learners invoke particle movement/mobility as the mechanism for conductivity, or merely name bond type; (b) whether the word 'structure' is used in a meaningful way or just inserted to comply. Mentally note two learners who show mechanistic thinking and two who show surface-level thinking — use these four as cold-call targets in Phase 3.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive an A3 'Evidence Organiser' sheet pre-printed with a 3×5 table: rows labelled NaCl (table salt), Al (aluminium sufuria wire), C (graphite pencil rod); columns labelled Bond Type, Particle-Level Structure (what is inside?), Conductivity Evidence (from our tests), Melting-Point Evidence (from our data), Mechanical Behaviour Evidence (from our tests). Groups spend 12 minutes mining their science notebooks from Lessons 1–6 to fill every cell with specific evidence, not general statements. They must write the actual numbers where available (e.g. 'NaCl melting point: 801 °C	Distribute A3 Evidence Organiser sheets and say: 'Every single cell must contain evidence from a specific lesson — not something you think you remember from primary school, not something from Google. Lesson number and observation. You have 12 minutes.' Set a visible countdown timer. Circulate continuously. At the 5-minute mark, pause the class for 30 seconds and say: 'Raise your hand if your Melting Point Evidence column is empty for ANY substance.' Address those groups directly: 'Go back to your Lesson 3 notebook page — the simulation data table.' At the 10-minute mark, cold-call ONE group	Evidence Mining with Role-Differentiated Accountability. Assigning the Evidence Checker and Vocabulary Guard roles ensures learners engage with prior-lesson records rather than relying on vague memory, and that scientific vocabulary is deployed accurately. The A3 organiser makes the evidence structure visible and portable — groups will carry it into the gallery walk in Phase 3.	Photograph two or three completed Evidence Organiser sheets with a phone or tablet for later reference. Check specifically: (a) Are conductivity entries distinguishing between solid-state and dissolved/molten NaCl? This is the most common omission. (b) Do Al entries mention delocalised electrons? (c) Do graphite entries mention layered structure and mobile π-electrons? Flag groups that omit these for targeted questioning in Phase 3.

	measured in simulation; Al melting point: 660 °C from data table'). One group member is the 'Evidence Checker': they must challenge any cell entry that cannot be traced to a specific lesson observation. A second member is the 'Vocabulary Guard': they ensure the terms ionic, metallic, giant covalent/atomic, delocalised electrons, lattice, and layers appear at least once across the table.	to read out their Al evidence row. Apply 10-second WAIT TIME, then ask the class: 'Does anyone want to add to or challenge what they just said?' Take 2–3 responses. Do NOT yet discuss the Explain column — that comes in Phase 3.		
Explain Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Using their completed Evidence Organiser, each group now writes three CER (Claim–Evidence–Reasoning) explanations — one for each substance — on large sheets of paper (A2 or flipchart paper) to be posted on the wall. The CER template is on the board: CLAIM: '[Substance] has [property] because of its [bond/structure type].' EVIDENCE: 'In Lesson [X] we observed/measured [specific data point].' REASONING: 'This happens because at the particle level, [describe what the particles/electrons are doing].' Groups have 18 minutes. After posting, the class conducts a 10-minute silent Gallery Walk: each learner carries two sticky notes — one green ('I agree and can add'), one red ('I challenge this'). They read all posted explanations and place sticky notes with written comments. Groups then return to their own poster and read the feedback (3 minutes). A 10-minute whole-class debrief follows: the teacher selects the three most contested red sticky notes and facilitates discussion.	Introduce CER: 'Scientists do not just state facts — they make a claim, back it with evidence, and explain the mechanism. That is what you are doing now.' Display the template. During the 18-minute writing period, use the 'Press for Mechanism' move: when you see a reasoning sentence that describes without explaining (e.g. 'because it is ionic'), crouch to group level and say quietly: 'What are the ions actually DOING — or NOT doing — in the solid that stops charge from flowing? Rewrite that sentence.' Apply this move to at least THREE groups. During the Gallery Walk, enforce silence: 'No talking — let the posters speak.' During the debrief, for each contested red sticky note read it aloud and cold-call the group that wrote the original claim: 'Do you stand by this claim? What is your evidence?' Allow 10-second WAIT TIME. Then cold-call a challenger: 'What is your alternative reasoning?' Facilitate, do not resolve — push learners to resolve using evidence. Close by saying: 'Let us build the consensus explanation together.' On the board, write the three agreed mechanistic explanations in full	CER Framework combined with Gallery Walk Peer Critique. The CER structure forces learners to separate observation from interpretation, and to articulate mechanism rather than merely label bond type. The Gallery Walk externalises reasoning for peer scrutiny, making thinking visible and creating low-stakes accountability. The 'Press for Mechanism' teacher move is the key pedagogical lever of this lesson: it shifts explanations from 'what' to 'why', which is the core sensemaking goal of the Explain phase.	Collect all three CER posters at the end of the lesson. Assess against three criteria: (1) Is the CLAIM correctly linked to bond/structure type? (2) Does the EVIDENCE cite a specific, lesson-sourced data point? (3) Does the REASONING describe particle or electron behaviour (not just bond-type name)? Use this assessment to identify learners who need targeted support before Lesson 8 (the Model Building finale). During the Gallery Walk, note which misconceptions appear on multiple red sticky notes — these will drive the Lesson 8 opening review.

		sentences, one per substance, crowd-sourced from the class.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board (DQB) posted at the front of the room — a large sheet of paper or board section that has been accumulating questions and partial answers since Lesson 1. Working as a whole class (7 minutes), learners nominate DQB entries that can now be ANSWERED based on today's explanations. Nominated entries are moved from the 'Still wondering' column to the 'We can now explain' column, and a learner writes a one-sentence answer beside each moved entry. Learners also nominate any NEW questions raised by today's lesson. The teacher then poses the Transfer Prediction task (5 minutes, individual written response in notebooks): 'Iodine crystals are used in water-purification tablets sold at pharmacies in Nairobi — iodine (I_2) forms simple molecular crystals. Iron is used in Jua Kali welding rods. Using ONLY what you now know about structural types, predict: (a) Will iodine have a high or low melting point? (b) Will solid iodine conduct electricity? (c) Will iron conduct electricity as a solid? Justify each prediction in one sentence.'	Stand at the DQB and say: 'Look at every question on this board. Which ones can we now answer — fully, with evidence and mechanism?' Cold-call FIVE different learners (choose learners who have not yet spoken in today's lesson). For each answer, ask the class: 'Thumbs up if you agree with this answer as stated — thumbs sideways if you think it needs more precision.' Revise answers live based on class input. For the Transfer Prediction task, say: 'Close all notes. This is you applying the framework — not recalling facts.' Allow 5 minutes of silent individual writing. Do NOT discuss answers yet — say: 'We will test these predictions at the start of Lesson 8.' Collect notebooks or ask learners to photograph their predictions so they cannot be altered.	DQB Curation with Transfer Prediction. Moving questions from 'still wondering' to 'we can explain' makes conceptual progress visible and gives learners a sense of achievement. The Transfer Prediction task is a formative transfer probe: it tests whether learners can apply the structural framework to novel substances without procedural scaffolding. Restricting it to unfamiliar Kenyan-context substances (iodine tablets, welding iron) prevents pattern-matching from memory.	Collect or photograph Transfer Prediction notebook entries. Mark against a simple three-point rubric per substance: 0 = no justification; 1 = justification names structural type only; 2 = justification names structural type AND states what the particles/electrons do. A score of less than 4/6 across all three predictions flags a learner for targeted support. Use this data to seat learners strategically for the Lesson 8 model-building activity.
Model Building Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners open their cumulative particle model — the annotated diagram they have been building since Lesson 1 — and spend 10 minutes making specific, labelled revisions. Today's required revisions are displayed on the board: (R1) Add a caption to the NaCl section: 'In solid NaCl, ions	Display the four required revisions on the board and say: 'Your model has been growing for six lessons. Today you write the explanatory captions that turn a picture into a scientific explanation. Every caption must answer WHY, not just WHAT.' Circulate for 8 minutes. Look specifically for: (a)	Cumulative Model Revision with Explanatory Captioning. Adding particle-level captions transforms the model from a structural diagram into a mechanistic explanation tool. Requiring learners to paraphrase rather than copy ensures genuine meaning-making rather than transcription.	Conduct a 60-second 'Model Snapshot' at lesson end: ask every learner to hold up their model open to today's revisions. Scan the room for completeness of all four required captions. Note learners with incomplete or missing captions for a brief check-in at the start of Lesson 8. Look for the

for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	are held in fixed lattice positions → no mobile charge carriers → no conductivity. When dissolved or melted, ions become mobile → conducts.' (R2) Add a caption to the Al section: 'Sea of delocalised electrons moves freely through fixed metal ion lattice → conducts in solid state; lattice layers slide over each other → malleable.' (R3) Add a caption to the graphite section: 'Each carbon bonded to 3 others in flat hexagonal layers; one delocalised electron per carbon free to move along layers → conducts. Weak Van der Waals forces between layers → layers slide off easily (pencil marks on paper).' (R4) Add a new 'Property Prediction Key' box: a colour-coded legend linking each structural type to its typical property profile. Learners who finish early add a fourth mini-diagram for diamond, labelling why its properties differ from graphite despite both being carbon.	whether learners are copying captions verbatim from the board, or paraphrasing in their own words — if copying, say: 'Put the board caption in your own words — what does that mean to you?'; (b) whether the Property Prediction Key uses correct colour-coding without errors. In the final 2 minutes, cold-call THREE learners to read one of their captions aloud. After each reading, ask: 'Did that caption say WHY? Give them a thumbs up if it did.' Close the lesson by returning to the anchoring phenomenon: point to the three objects (or their images) and say: 'At the start of this unit, these three objects were mysterious. Today you explained all three of them — from atoms to sufuria. That is chemistry. Next lesson, we build the final model and bring it all together.'	The closing return to the anchoring phenomenon re-anchors all abstract knowledge to the original concrete puzzle, reinforcing the coherence of the unit storyline.	specific vocabulary 'delocalised electrons', 'lattice', 'mobile ions' and 'Van der Waals' in the captions — absence of these terms in a caption that is otherwise correct indicates surface-level understanding that needs consolidation.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. MECHANISM VS. LABEL: When you applied the 'Press for Mechanism' move, how many groups shifted from labelling ('it is ionic') to genuine mechanistic reasoning ('the ions cannot move in the solid')? Which groups still need support before Lesson 8?

2. DQB QUALITY: Were the answers placed in the 'We can now explain' column genuinely mechanistic, or did learners accept surface-level answers too quickly? If the latter, how will you challenge this in Lesson 8?

3. TRANSFER PREDICTIONS: What proportion of learners scored 4 or more out of 6 on the iodine/iron transfer task? If fewer than 70% reached this threshold, what targeted review will you build into the first 10 minutes of Lesson 8?

4. GRAPHITE MISCONCEPTION: Did any learners still hold the misconception that graphite conducts 'because it is a metal' or 'because it is shiny'? How was this addressed today, and how will you consolidate the correct explanation in Lesson 8?

5. KENYAN CONTEXT CONNECTION: Did learners make spontaneous connections between the structural explanations and their own experience (e.g. why the sufuria handle bends but does not snap; why a pencil leaves a mark; why dissolving salt in water before a conductivity test changes the result)? Note these connections — they are evidence of transfer and can be used as opening hooks in Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We tested and recorded data on the conductivity, melting points and mechanical behaviour of table salt ($NaCl$ from the Kenyan coast), aluminium (from a sufuria handle) and graphite (from a used AA battery rod) across Lessons 1–6. We observed that dry salt does not conduct but salt solution does; aluminium conducts as a solid and bends without breaking; graphite conducts as a solid but snaps rather than bends, and leaves marks on paper.
What did I learn?	Each substance has a different structural type determined by its bond type: $NaCl$ forms a giant ionic lattice where ions are fixed in position (so no conductivity in solid state, but high melting point); aluminium forms a giant metallic lattice with a sea of delocalised electrons (so conducts in solid state, is malleable, and has a moderate melting point); graphite forms a giant covalent layered structure where each carbon has one delocalised electron free to move along the layer (so conducts), and weak Van der Waals forces between layers mean layers slide off easily (so it is brittle across layers and marks paper). We can predict the properties of any substance if we know its structural type.
How does this explain the phenomenon?	The three kitchen items look similar as solids but behave so differently because their internal particle-level structures — determined by their bond types — are completely different. In $NaCl$, charged ions locked in a rigid lattice cannot move to carry current (hence no conductivity in solid form) but the strong electrostatic forces holding the lattice together require enormous energy to overcome (hence a very high melting point of 801 °C). In aluminium, metallic bonding produces a 'sea' of freely moving delocalised electrons that carry charge immediately (hence solid conductivity) and a lattice of positive ions whose layers can slide (hence malleability). In graphite, covalent bonding within layers with one free electron per carbon enables conductivity along the layers, while weak Van der Waals forces between layers allow them to shear apart onto paper. The bond type controls the structure; the structure controls every property we can measure.

LESSON 8 (80 minutes): Final Explanation — Four Structural Types, Jigsaw Presentations and the Complete Story of the Kitchen Items

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To synthesise all sub-strand learning by classifying NaCl, aluminium and graphite into their giant structural categories and constructing a complete, evidence-based explanation of every observed property difference from the anchoring phenomenon.
Knowledge	Learners will be able to: (a) classify substances into four structural types — giant ionic, simple molecular, giant atomic and giant metallic; (b) describe the particle arrangement, bonding forces and properties of each structural type; (c) explain the properties of NaCl (giant ionic), iodine (simple molecular), diamond and graphite (giant atomic/layered covalent) and aluminium (giant metallic); (d) describe the relationship between bond types and physical properties of elements and compounds.
Skills	Learners will be able to: (a) use print and digital media to research structural types and record findings; (b) sketch and label structural diagrams of the four structural types; (c) prepare and deliver a concise jigsaw explanation to peers; (d) evaluate and revise a cumulative particle model built across all eight lessons; (e) select appropriate materials for use based on bond type and use.
Attitudes	Learners will: (a) appreciate that understanding particle-level structure enables informed, evidence-based decisions about material selection in everyday Kenyan life; (b) reflect on the appropriateness of substances used in day-to-day life; (c) value collaboration, peer teaching and respectful scientific discussion.
Key Inquiry Question	How do bond types relate to physical properties of compounds?
Purpose in Storyline	This is the culminating lesson of Sub-Strand 2.3. All threads converge here: ionic bonding (Lessons 2–3), covalent bonding (Lessons 4–5), metallic bonding (Lesson 6) and Van der Waals / hydrogen bonding (Lesson 7) are now formally unified under the four giant structural categories. Learners finally possess the complete vocabulary and conceptual framework to answer the driving question in full. The jigsaw strategy ensures every learner teaches as well as learns, and the final DQB review confirms that all opening questions have been resolved.
Safety Notes	No hazardous chemicals are used in this lesson. If physical samples of iodine crystals, NaCl, aluminium foil or graphite rods are displayed for reference, learners should not handle iodine crystals directly — teacher holds the sealed sample vial. Remind learners of screen-time hygiene when using digital devices for research. Remind learners not to ingest any substances even if they appear familiar (e.g. table salt samples used in class).

B. LESSON OVERVIEW

Lesson 8 is the final-explanation lesson for Sub-Strand 2.3. The teacher opens by briefly returning to the anchoring phenomenon — the aluminium sufuria handle, the graphite rod and the table salt on the kitchen table — and challenges learners to confirm how much of the driving question they can now answer. Learners are organised into four expert groups (giant ionic, simple molecular, giant atomic, giant metallic) and spend 15 minutes researching their assigned structural type using print or digital media, then sketching a labelled structural diagram. Groups then jigsaw into mixed teams of four (one expert per structural type) and each expert delivers a 2-minute explanation while teammates record key features on a summary table. A whole-class gallery of structural diagrams is assembled on the board. The teacher then leads a guided synthesis in which the three kitchen items are explicitly mapped onto the framework, linking every property observed in Lesson 1 to the bond type and structural category identified across Lessons 2–7. Learners revise their particle models for the final time (L1 sketch vs. final model comparison). The DQB is reviewed and closed — every sticky question is answered. The lesson closes with a teacher-led reflection on how this knowledge supports real-world material choices (e.g. why aluminium is used for sufurias, why graphite is used in dry-cell battery electrodes, why NaCl is safe to dissolve in food but requires high heat to melt).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners re-read the anchoring phenomenon description (displayed on the board or read aloud by the teacher). Working individually for 3 minutes, they write a 'Final Prediction Statement' in their notebooks answering: 'Using the bond types and structures you have studied, write one sentence each to explain (a) why dry NaCl does not conduct electricity but its solution does; (b) why aluminium bends without breaking; and (c) why graphite conducts electricity even though it is a non-metal.' Learners then share with a shoulder partner for 2 minutes before two or three responses are cold-called.	Teacher re-displays or re-reads the anchoring phenomenon slowly and deliberately. Says: 'Before we do anything else today, I want you to write what YOU now think the complete answer is. Use every term you have learned — ionic lattice, delocalised electrons, layers, Van der Waals, metallic bonding. You have three minutes — go.' [WAIT TIME: 3 minutes silent individual writing.] After partner share, cold-calls three learners using name cards: 'Amina, read me your sentence about NaCl. Baraka, what did you write about aluminium? Zawadi, tell us about graphite.' Teacher scribes key phrases on the board WITHOUT correcting yet — these become the 'entry benchmark' to compare with the final explanation at the end of the lesson.	Retrieval Practice / Benchmark Writing — learners activate and externalise all prior learning before new instruction. Recording the entry benchmark on the board creates a visible before-and-after comparison that makes conceptual growth concrete.	Teacher listens for: correct use of 'delocalised electrons' for both aluminium and graphite; correct use of 'mobile ions' for NaCl in solution vs. 'fixed lattice ions' for dry NaCl. Absence of these terms flags gaps to address in Phase 3. Teacher mentally notes which learners used only surface-level language ('because it is a metal / because it dissolves') to target in cold-calls during synthesis.
Observe Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are pre-assigned to one of four expert groups (6–7 learners each in a class of 25–28): Group A — Giant Ionic (NaCl); Group B — Simple Molecular (iodine, I_2); Group C — Giant Atomic (diamond and graphite); Group D — Giant Metallic (aluminium). Each group receives a research brief card listing: the structural type name, the substance(s) to research, four specific questions to answer (particle identity, type of force holding particles, how the structure explains electrical conductivity, how the structure explains melting point), and a blank diagram template. Groups use available	Teacher circulates every 2–3 minutes with targeted prompts: To Group A (ionic): 'Show me on your diagram WHERE the Na^+ and Cl^- ions sit — are they in rows? What holds them in place?' To Group B (simple molecular): 'Iodine is a solid at room temperature but melts easily at 114 °C — what does that tell you about the forces BETWEEN molecules?' To Group C (giant atomic): 'You have TWO substances here. How is graphite's structure DIFFERENT from diamond's, and which difference explains why graphite conducts electricity?' To Group D (metallic): 'Show me the sea of electrons on	Jigsaw Expert Group Research (Aronson Jigsaw Protocol) — dividing the four structural types across groups ensures that every learner develops genuine expertise in one area and must teach it to others, creating interdependence and distributed accountability. Sketching labelled diagrams (NGSS SEP 2: Developing and Using Models) converts verbal descriptions into spatial representations that reveal structural logic.	Teacher checks diagrams for: (A) ionic — 3D lattice with alternating Na^+ and Cl^-, label 'electrostatic attraction'; (B) molecular — discrete I_2 molecules with intramolecular covalent bond and intermolecular Van der Waals forces labelled separately; (C) diamond — tetrahedral covalent network vs. graphite — hexagonal layers with labelled delocalised electrons and Van der Waals between layers; (D) metallic — regular cation lattice in a 'sea' of delocalised electrons. Any diagram missing the force labels is flagged for correction before jigsaw.

	textbooks, printed reference sheets or digital devices for 15 minutes to research and sketch a fully labelled structural diagram. A physical sample or photograph of their substance is placed on each group's table for reference (sealed iodine vial for Group B; NaCl crystals, aluminium foil and graphite rod for others).	your diagram. Which part of the diagram explains why aluminium can be drawn into wires without snapping?' At the 10-minute mark teacher gives a 5-minute warning: 'Five minutes — make sure every particle is labelled and every force is named on your diagram.' At 15 minutes: 'Pencils down. Appoint one speaker and one diagram-holder for the jigsaw.'		
Explain Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Expert groups dissolve and learners regroup into mixed jigsaw teams of four (one A, one B, one C, one D expert per team). Each expert delivers a 2-minute explanation of their structural type while teammates complete a four-column summary table in their notebooks (columns: Structural Type | Particles Present | Forces Holding Structure | Key Properties Explained). After all four experts have shared (8 minutes total), the teacher leads a 12-minute whole-class synthesis. The class then works in pairs to write a 'Final Explanation Paragraph' answering the driving question fully, explicitly naming all three kitchen items, their structural type and the bond type responsible for each observed property. Pairs read their paragraphs aloud and the class votes on the most complete explanation using a thumbs scale.	Before jigsaw sharing begins, teacher says: 'You are not just sharing — you are TEACHING. Your three teammates will only know your structural type from what you say and show. Make it clear.' [Sets 2-minute timer per expert.] During sharing, teacher circulates and listens. After all four experts have shared, teacher returns to the front and says: 'Now let us bring all four structures back to our three kitchen items.' Teacher draws a three-column table on the board (NaCl | Aluminium | Graphite) and cold-calls: 'Group A expert — which structural type is NaCl? Tell us one property from the phenomenon and explain it using the lattice.' [WAIT TIME: 10 seconds.] 'Group D expert — aluminium. Baraka, why does it bend?' [WAIT TIME: 10 seconds.] 'Group C expert — graphite. Zawadi, why does the graphite rod conduct electricity in the circuit even though carbon is a non-metal?' [WAIT TIME: 12 seconds.] Teacher scribes answers into the table. Then: 'Now look at the board from Lesson 1 — the questions we could NOT answer. Look at our DQB. Let us go through every sticky note.' Teacher removes each resolved question from the DQB as the	Jigsaw Peer Teaching followed by Driving Question Synthesis — peer teaching leverages the protégé effect (learners understand material more deeply when they must explain it). The explicit DQB review (NGSS SEP 6: Constructing Explanations) transforms accumulated evidence into a coherent, publicly verified answer to the driving question. Writing the Final Explanation Paragraph consolidates understanding through scientific argumentation (NGSS SEP 7).	Teacher assesses Final Explanation Paragraphs for four non-negotiable elements: (1) all three kitchen items are named and assigned a correct structural type; (2) at least one specific force is named for each structure (electrostatic attraction / delocalised electrons / covalent network / Van der Waals between layers); (3) conductivity is explained in terms of particle mobility (ions or electrons); (4) melting point is explained in terms of force strength. Pairs whose paragraphs omit any element are asked a targeted follow-up question before the lesson ends.

		class confirms the answer. For the Final Explanation Paragraph, teacher says: 'With your partner, write a paragraph that a Form 1 student could read and understand — use ALL four structural types, name the forces, name the properties. You have 6 minutes.' Cold-calls two pairs to read. Scribes the class's agreed 'best answer' on the board in green marker.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The full DQB (assembled across Lessons 1–7) is displayed at the front. Learners re-read every sticky note silently for 2 minutes. Individually, each learner writes in their notebook: 'ANSWERED' or 'STILL UNCLEAR' next to a numbered list of the DQB questions they copied in Lesson 1. The class then votes on each DQB question: teacher reads it aloud and asks for a show of hands — 'Who can answer this now?' If consensus is reached, the sticky note is moved to the 'ANSWERED' column. Any question that remains in the 'STILL UNCLEAR' column is addressed by the teacher with a brief 60-second explanation. Learners then write one final 'New Wondering' — a question that the sub-strand has raised for them that goes beyond what was studied (e.g. 'Why does diamond conduct heat but not electricity if it has the same covalent network as graphite?').	Teacher reads each DQB sticky note aloud clearly: 'Our first question from Lesson 1 was — why does dry salt not conduct electricity? Hands up if you can answer that now.' [WAIT TIME: 5 seconds.] Cold-calls one learner for a full-sentence answer. Moves sticky note to 'ANSWERED' column with deliberate ceremony: 'We figured that out in Lesson 2 when we studied the ionic lattice — well done.' Continues for all remaining sticky notes. For any unresolved note, teacher gives a direct 60-second explanation and then moves it to 'ANSWERED.' At the end says: 'Every question we asked in Lesson 1 is now answered. Write your one New Wondering — this is what scientists do: every answer opens a new question. Share with the person next to you.' [WAIT TIME: 2 minutes writing + 1 minute sharing.] Cold-calls three 'New Wonders' and validates each: 'That is a university-level question — you are thinking like a materials scientist.'	DQB Closure Protocol — systematically resolving every opening question provides learners with a concrete record of their own conceptual growth across the sub-strand. The 'New Wondering' activity models authentic scientific practice (NGSS SEP 1: Asking Questions) and prevents the misconception that science provides final, closed answers.	Teacher scans learner notebooks to confirm that 'ANSWERED' is marked against all major DQB questions. The quality of 'New Wonderings' serves as a stretch indicator: surface-level new questions ('Why is salt white?') indicate partial understanding; questions about structure–property relationships at a deeper level ('Does every giant atomic solid conduct heat the same way?') indicate secure understanding.
Model Building Phase 📹 **VIDEO:**	Learners retrieve the particle sketch they drew in Lesson 1 (kept in their notebooks or portfolios) showing their initial ideas about	Teacher says: 'Turn to your Lesson 1 sketch — the one you drew when we first looked at the salt, the sufuria handle and the	Comparative Model Revision (NGSS SEP 2: Developing and Using Models) — placing the initial model alongside the final model	Teacher collects (or photographs) Final Model diagrams. Looks for: (1) NaCl — alternating Na^{+}/Cl^{-} in a lattice, label 'electrostatic

Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	what was happening inside NaCl, aluminium and graphite. They place this sketch next to a new 'Final Model' page and spend 8 minutes drawing three final, fully labelled particle diagrams — one for each kitchen item — showing the correct structural type, particle identity, bonding forces and the feature that explains conductivity or lack thereof. They then annotate the Lesson 1 sketch with red pen, circling every error or missing idea and writing a brief note explaining what they now know that they did not know in Lesson 1. A 3-minute whole-class share follows where two or three learners hold up both pages and describe their biggest conceptual change. The lesson closes with the teacher re-reading the anchoring phenomenon aloud for the final time, pausing after each observation for the class to call out the explanation in unison.	graphite rod. Look at it carefully. In a moment you will draw your FINAL model — everything you now know.' [Gives 8 minutes.] Circulates and looks for: correct 3D ionic lattice for NaCl, delocalised electron sea for aluminium, hexagonal layer structure for graphite with labelled Van der Waals between layers and delocalised electrons within layers. After 8 minutes: 'Now take a red pen. Go back to your Lesson 1 sketch. Circle every mistake — be kind to your past self, but be honest. Write next to each circle what you know NOW.' [3 minutes.] Cold-calls two learners: 'Amina, show us your Lesson 1 sketch and tell us the biggest thing you got wrong. What do you know now?' After sharing, teacher re-reads the phenomenon slowly: 'The salt dissolves and the solution conducts — ' [pauses, class calls out: 'because the ions become mobile in solution!']. 'The aluminium bends without breaking — ' [class: 'because the cation layers slide over the sea of delocalised electrons!']. 'The graphite conducts electricity even though carbon is a non-metal — ' [class: 'because delocalised electrons move freely within the hexagonal layers!']. Teacher concludes: 'You have just answered the question we started with eight lessons ago. That is what science learning looks like.'	makes the trajectory of learning visible and explicit. The red-pen annotation ritual is a metacognitive tool that encourages learners to own their prior misconceptions without shame. The choral phenomenon re-reading ritual creates a shared, celebratory consolidation of the entire sub-strand's core explanation.	attraction,' note 'ions fixed in solid → no conductivity; mobile in solution → conducts'; (2) Aluminium — regular Al^{3+} cation lattice, 'sea of delocalised electrons,' notes 'electrons carry charge → conducts as solid; layers slide → malleable'; (3) Graphite — hexagonal carbon layers, 'covalent bonds within layer,' 'Van der Waals between layers,' 'delocalised electrons within layer → conducts; weak interlayer forces → layers flake off → pencil mark on paper.' Diagrams missing any of these specific labels are flagged for individual follow-up.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal:

1. DRIVING QUESTION RESOLUTION: When the class called out the choral answers during the final phenomenon re-read, were all three explanations (NaCl, aluminium, graphite) confidently and correctly stated? If any kitchen item's explanation was hesitant or incorrect, which structural concept still needs consolidation — and how will you address it at the start of Sub-Strand 2.4? 2. JIGSAW EQUITY: Did all four expert groups produce diagrams of comparable quality before the jigsaw? If Groups B (simple molecular) or C (giant atomic) produced weaker diagrams, consider whether the research brief card for those groups needs more scaffolding — e.g. a partially completed diagram template rather than a blank one. 3. FINAL EXPLANATION PARAGRAPHS: Review 3–4 sampled paragraphs. Did learners correctly distinguish between intramolecular forces (covalent bonds within graphite layers) and intermolecular forces (Van der Waals between layers)? This is the most common conflation error in this sub-strand. If present, plan a 5-minute clarification during the next lesson's warm-up. 4. DQB COMPLETION: Were there any sticky-note questions that no learner could answer and that required teacher explanation? Record these as diagnostic data — they represent concepts that were not sufficiently consolidated in earlier lessons and should inform your review planning before the summative assessment. 5. MODEL COMPARISON IMPACT: Did the Lesson 1 vs. Final Model comparison generate genuine surprise or pride in learners? Note which learners showed the most significant conceptual growth — these learners are strong candidates for peer-support roles in Sub-Strand 2.4. 6. KENYA CONTEXT: Did you successfully connect the structural type framework to locally relevant uses — aluminium sufurias, graphite in dry-cell batteries from Kenyan households, NaCl from the Kenyan coast? Note any additional local examples that learners themselves raised, as these can enrich future iterations of this lesson. 7. TIMING: This lesson is designed for 80 minutes. If the jigsaw sharing or the Final Explanation Paragraph activity ran over time, consider whether the expert-group research phase (15 minutes) could be assigned as a pre-lesson homework task in future cohorts, freeing lesson time for deeper synthesis discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In Lesson 1 we observed three items from a Kenyan kitchen: a table-salt crystal (NaCl from the coast), an aluminium sufuria handle wire and a graphite rod from a dry-cell battery. We saw that dry NaCl did not conduct electricity but its solution did and it had a very high melting point; aluminium conducted electricity immediately as a solid, was shiny and bent without breaking; and graphite — a non-metal — also conducted electricity yet was brittle and left layers on paper.
What did I learn?	Across Lessons 2–8 we learned that the differences between these three substances are explained by four giant structural types and their bonding forces: (1) Giant Ionic (NaCl) — Na^+ and Cl^- ions held in a 3D lattice by strong electrostatic attraction; ions are fixed in the solid so it cannot conduct, but dissolve to give mobile ions that do conduct; high lattice energy means a very high melting point. (2) Giant Metallic (Aluminium) — Al^{3+} cations in a regular lattice surrounded by a 'sea' of delocalised electrons; the electrons carry electrical charge so aluminium conducts as a solid; the cation layers can slide over the electron sea so the metal is malleable and ductile. (3) Giant Atomic / Layered Covalent (Graphite) — carbon atoms covalently bonded in flat hexagonal layers; each carbon uses only three of its four valence electrons for bonding, leaving one delocalised electron per atom that can carry charge along the layer — explaining why graphite conducts; the layers are held together only by weak Van der Waals forces, so they slide apart easily — explaining why graphite is brittle and leaves a pencil mark. We also studied simple molecular structures (e.g. iodine), where discrete molecules are held by weak Van der Waals forces, giving low melting points and poor conductivity.

How does this explain the phenomenon?	We can now give a complete, particle-level explanation of the anchoring phenomenon: the three kitchen items look similar as solids but behave so differently because they belong to different structural categories. NaCl is a giant ionic lattice — its high melting point and conductivity only in solution both follow directly from the strength and arrangement of electrostatic bonds between fixed ions. Aluminium is a giant metallic structure — its conductivity as a solid, its shine and its malleability all follow from the presence of freely moving delocalised electrons and the ability of cation layers to slide. Graphite is a giant atomic layered covalent structure — its unexpected conductivity follows from delocalised electrons within each hexagonal layer, while its brittleness and ability to leave marks on paper follow from the weak Van der Waals forces between layers. Bond type controls structure; structure controls every property we can measure.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.